Satellite Communications
ECE 4644 / 5610 Spring 2026

Week #12

April 6 - 10, 2026

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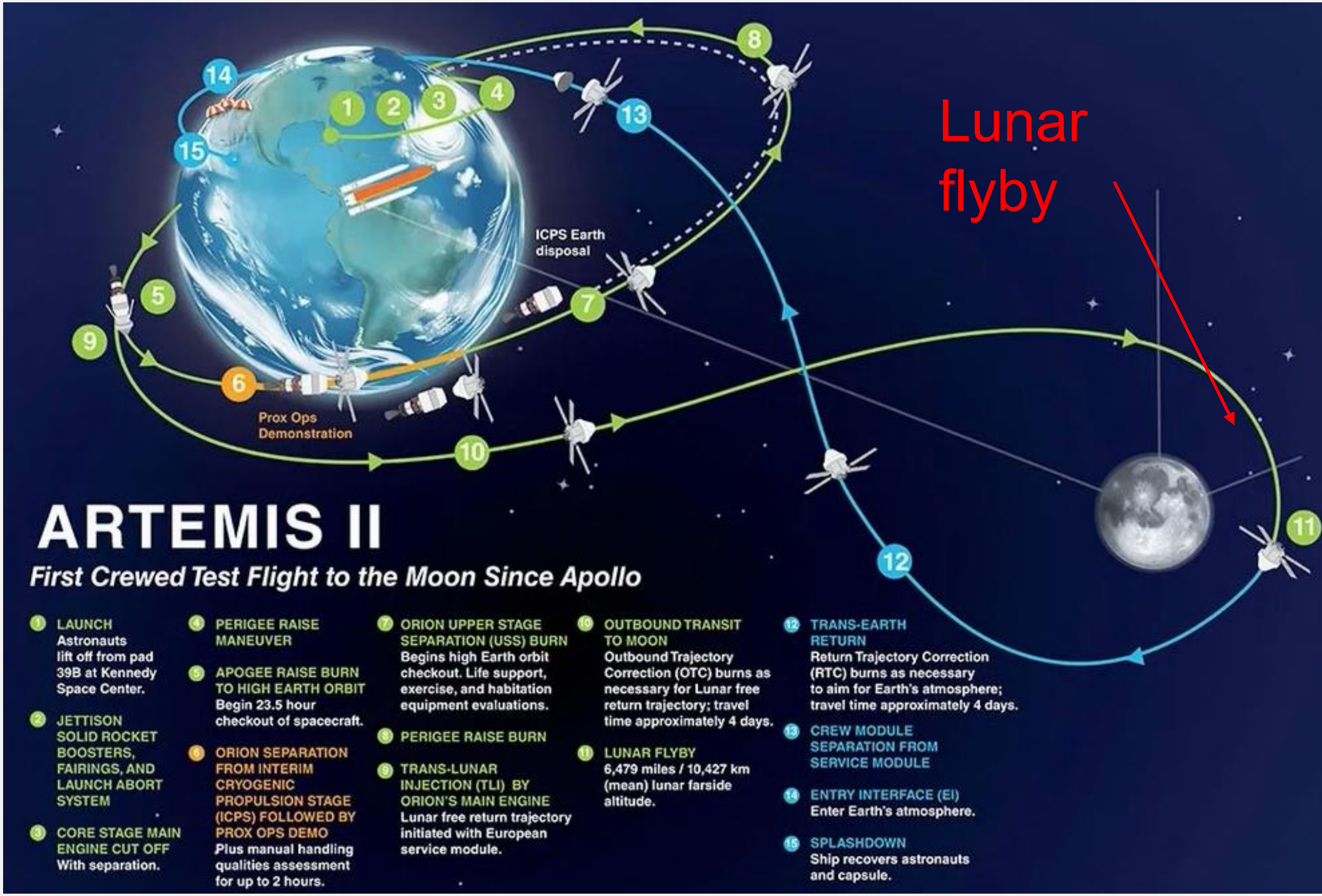
Topics (1/2)

- Artemis II update
- Digital phase modulation: BPSK and QPSK (Conclusion)
- Gray codes
- Bit error rate (BER)
- Multi-level signaling / APSK / QAM

Topics (2/2)

- Forward Error Correction (FEC)
- Parity bits / Block codes
- Convolutional coding
- Coding gain
- Burst errors / Interleaving
- Multiple Access (MA) techniques:
- FDMA
- TDMA

Update: Lunar flyby today (Monday) 2:45 – 9:20 PM



NASA's schedule for the lunar flyby

- Live coverage begins at 1 p.m. on Monday, April 6, and continues through 9:45 p.m.
- 1:30 p.m.: NASA hosts a conversation between the crew and the science officer
- 1:56 p.m.: The Artemis II crew is expected surpass the record previously set by the Apollo 13 crew in 1970 for the farthest humans have ever traveled from Earth.
- 2:45 p.m.: The seven-hour lunar observation period begins.
- 6:44 p.m.: Mission control expects to temporarily lose communication with the crew as Orion passes behind the Moon.
- 7:02 p.m.: Astronauts will make their closest approach to the Moon (4,070 miles), then reach its farthest point from Earth at 7:07 p.m.
- 7:25 p.m.: NASA's Mission Control Center should re-acquire communication with the astronauts.
- 8:35 p.m.: Orion enters period with Moon eclipsing the Sun and continues until 9:32 p.m.
- 9:20 p.m.: The flyby observation period wraps, and crew will begin transferring some of the imagery to the ground

QPSK Modulation - Review

- We can write QPSK waveform as

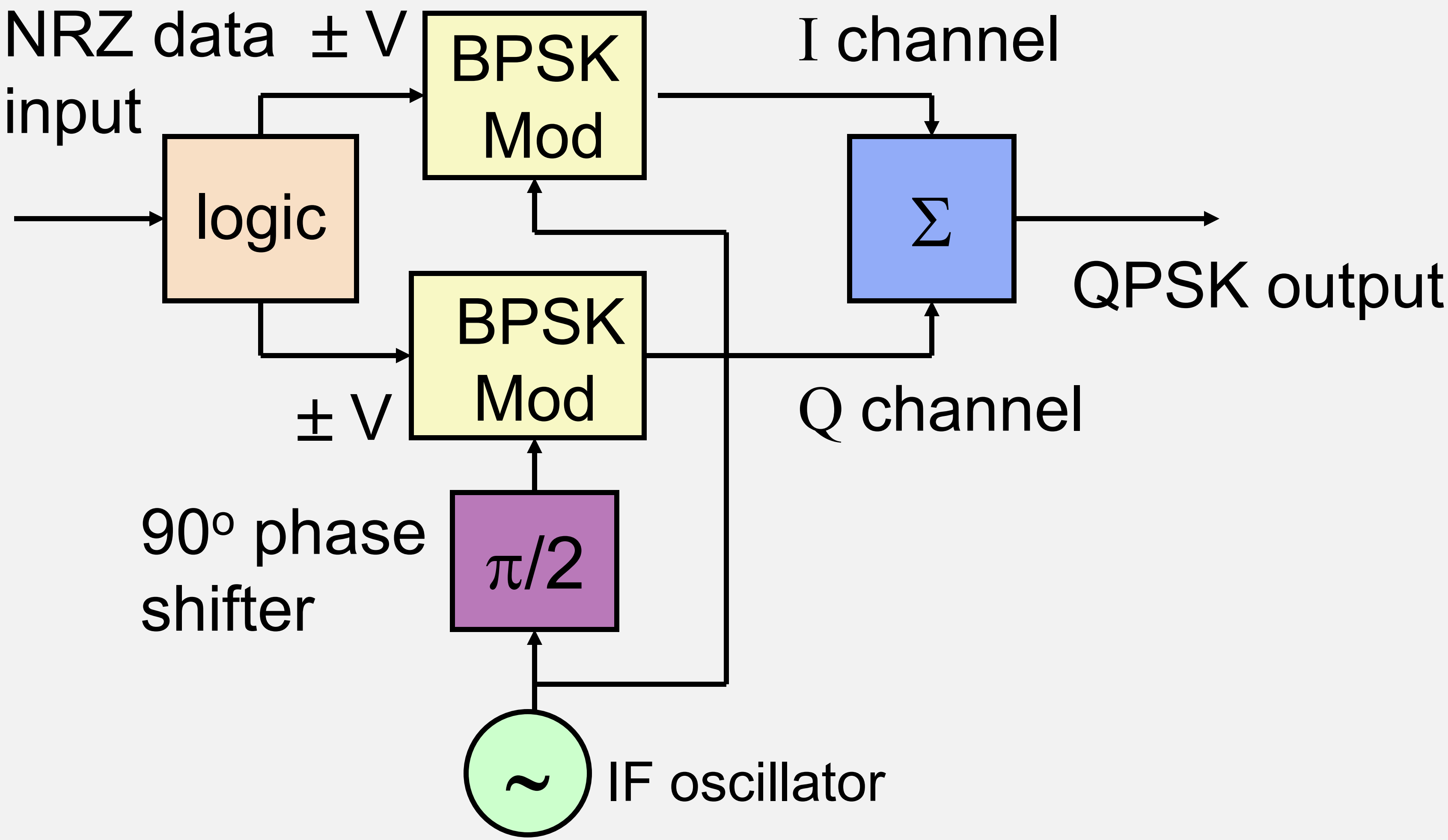
$$v_{\mathbf{QPSK}}(t) = U_{\mathbf{I}}(t)\, V \cos \omega t + U_{\mathbf{Q}}(t)\, V \sin \omega t$$

where

$$U_{\mathbf{I}} = \sqrt{2} \cos \phi \quad \text{and} \quad U_{\mathbf{Q}} = \sqrt{2} \sin \phi$$

are the modulating bits for the in-phase and quadrature channels (values: +1, -1)

QPSK Modulator



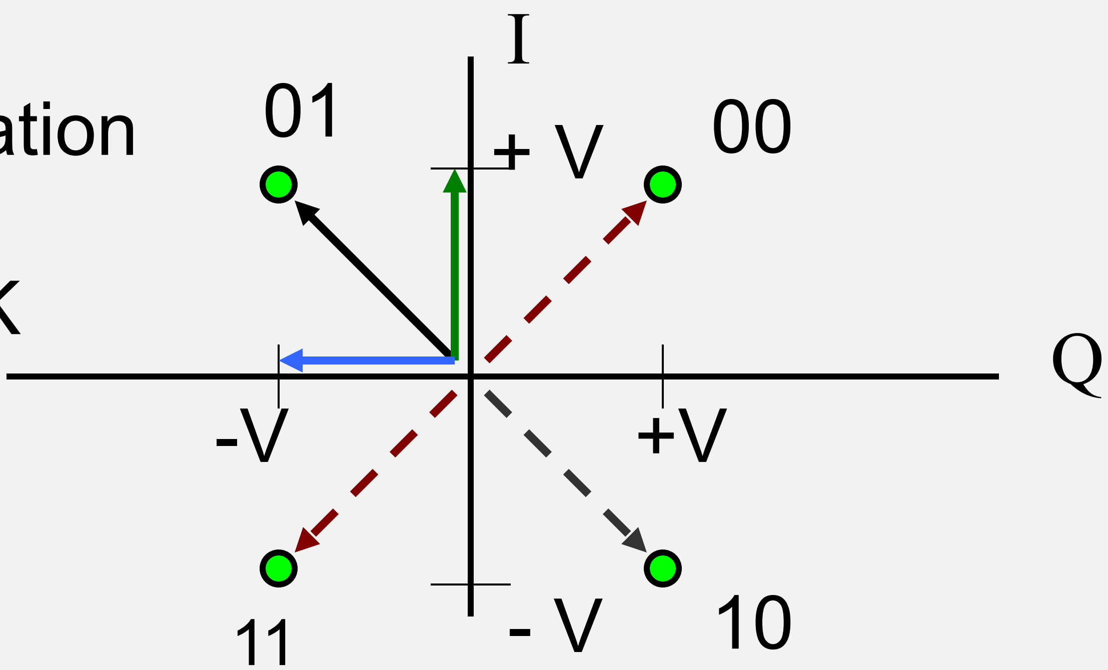
QPSK Modulator

- Input data 0 0 1 1 0 1 1 0
- I channel bits 0 1 0 1
- I Output (volts) +V -V +V -V
- Q channel 0 1 1 0
- Q Output (volts) +V -V -V +V
- QPSK Phase $\pi/4$ $5\pi/4$ $3\pi/4$ $7\pi/4$
- Can be seen from constellation diagram

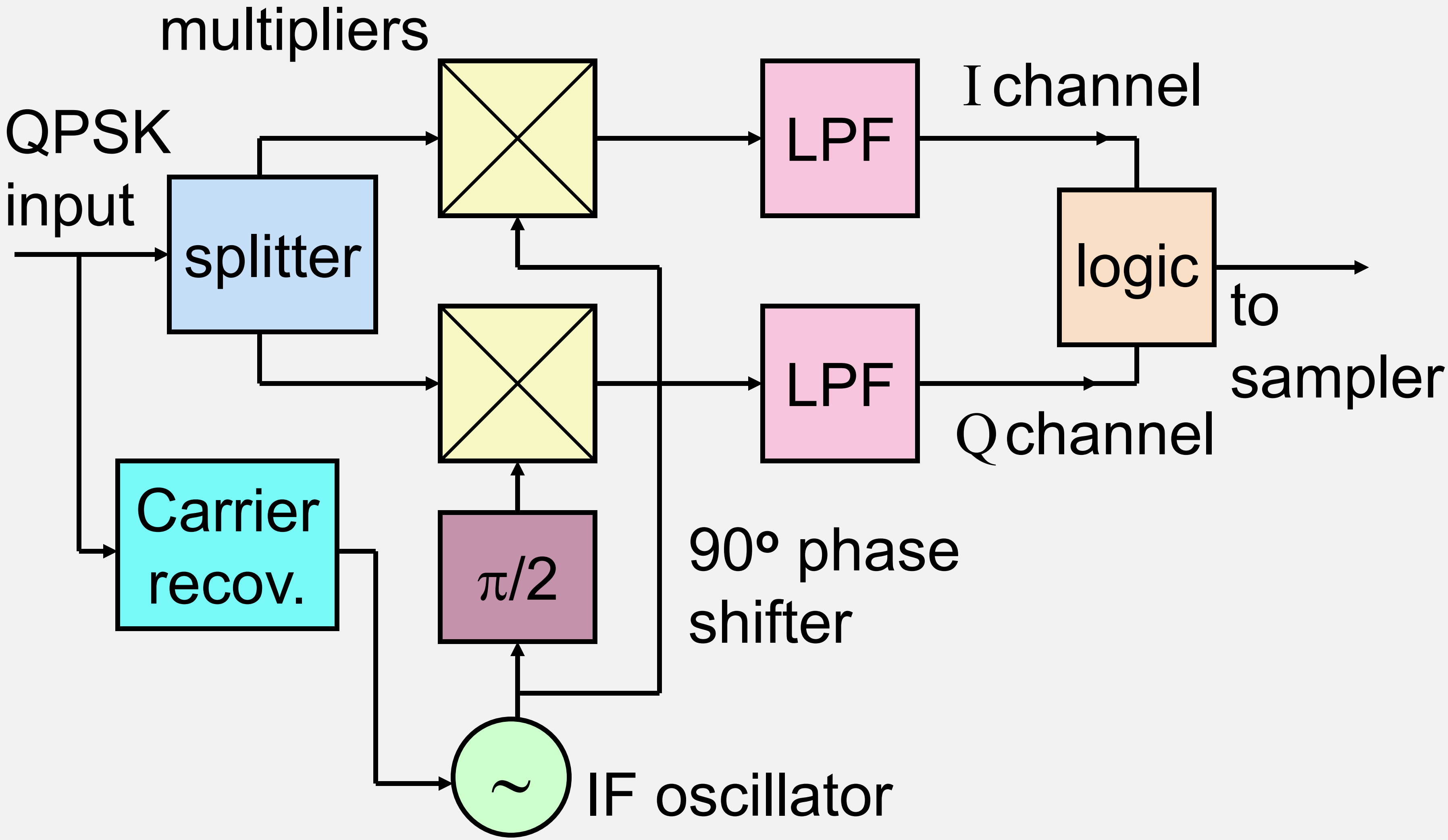
Constellation Diagrams

- Constellation diagram shows the tip of the I and Q phasor

Constellation Diagram for QPSK



QPSK Demodulator



QPSK Demodulation

- QPSK signal at IF stage is

  $v_{QPSK}(t) = U_I(t)\,V_{IF}\cos\omega t + U_Q(t)\,V_{IF}\sin\omega t$
- Apply same signal to two multipliers
- I channel multiplier is driven by in-phase LO

  $v_{I\,LO}(t) = \cos\omega t$
- Q channel multiplier driven by quadrature LO

  $v_{Q\,LO}(t) = \cos(\omega t + \pi/2) = \sin\omega t$

QPSK Demodulation

- I channel multiplier output is

$$v_I(t) = [U_I(t)\ V_{IF}\cos \omega t + U_Q(t)\ V_{IF}\sin \omega t] \times \mathbf{\cos \omega t}$$

$$= U_I(t)\ V_{IF} \cos \omega t \cos \omega t$$

$$+\ U_Q(t)\ V_{IF} \sin \omega t \cos \omega t$$

$$= U_I(t)\ V_{IF}\ \tfrac{1}{2}\ (1 + \cos 2\omega t)$$

$$+\ U_Q(t)\ V_{IF}\ \tfrac{1}{2} \sin 2\omega t$$

LPF removes double frequency terms

QPSK Demodulation

- LPF output of I channel is

$$v_I(t) = \tfrac{1}{2}\, U_I(t)\, V_{IF}$$

  This is the original I channel input $U_I(t)$ multiplied by ½ V_{IF}

- We have recovered $U_I(t)$

QPSK Demodulation

- Q channel multiplier output is

$$v_Q(t) = [U_I(t)\, V_{IF} \cos \omega t + U_Q(t)\, V_{IF} \sin \omega t] \text{ x } \mathbf{\sin \omega t}$$

$$= U_I(t)\, V_{IF} \cos \omega t \sin \omega t + U_Q(t)\, V_{IF} \sin \omega t \sin \omega t$$

$$= U_I(t)\, V_{IF}\, \tfrac{1}{2} \sin 2\omega t + U_Q(t)\, V_{IF}\, \tfrac{1}{2} (1 - \cos 2\omega t)$$

- I channel LPF removes double frequency terms

QPSK Demodulation

- LPF output of Q channel is

  $v_{\mathbf{I}}(t) = ½ U_{\mathbf{Q}}(t) V_{\mathbf{IF}}$

  This is the original I channel input $U_{\mathbf{Q}}(t)$ multiplied by $½ V_{\mathbf{IF}}$
- We have recovered $U_{\mathbf{Q}}(t)$
- Summing: we have recovered the values of the modulating bits (two at a time)

QPSK Modulation in practice

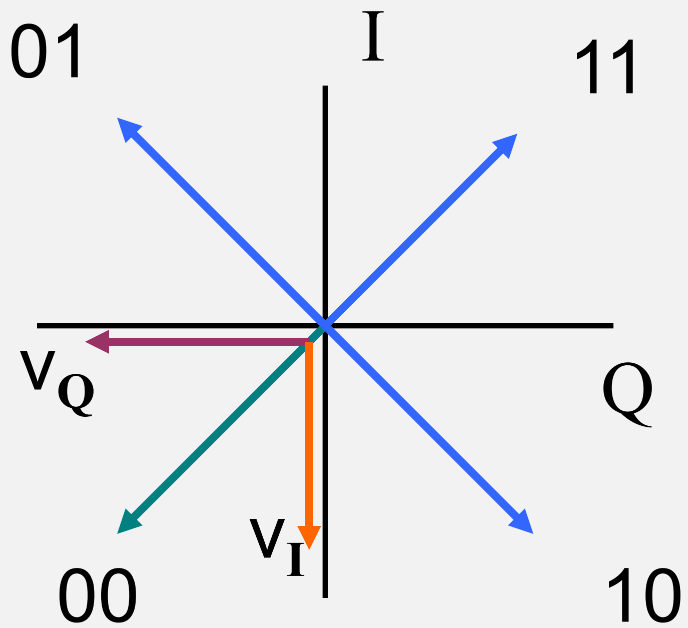
- QPSK signal is generated as sum of I and Q channel signals
- Logic circuit drives *both* I and Q modulators for each pair of input data bits
- E.g., 0 0 input drives I and Q channels with

  -1 signals and

$$v_{QPSK} = -\cos \omega t - \sin \omega t$$

$$= -\sqrt{2} \cos(\omega t + \pi/4)$$

QPSK generation in practice



Each phasor is generated by summing I and Q signals

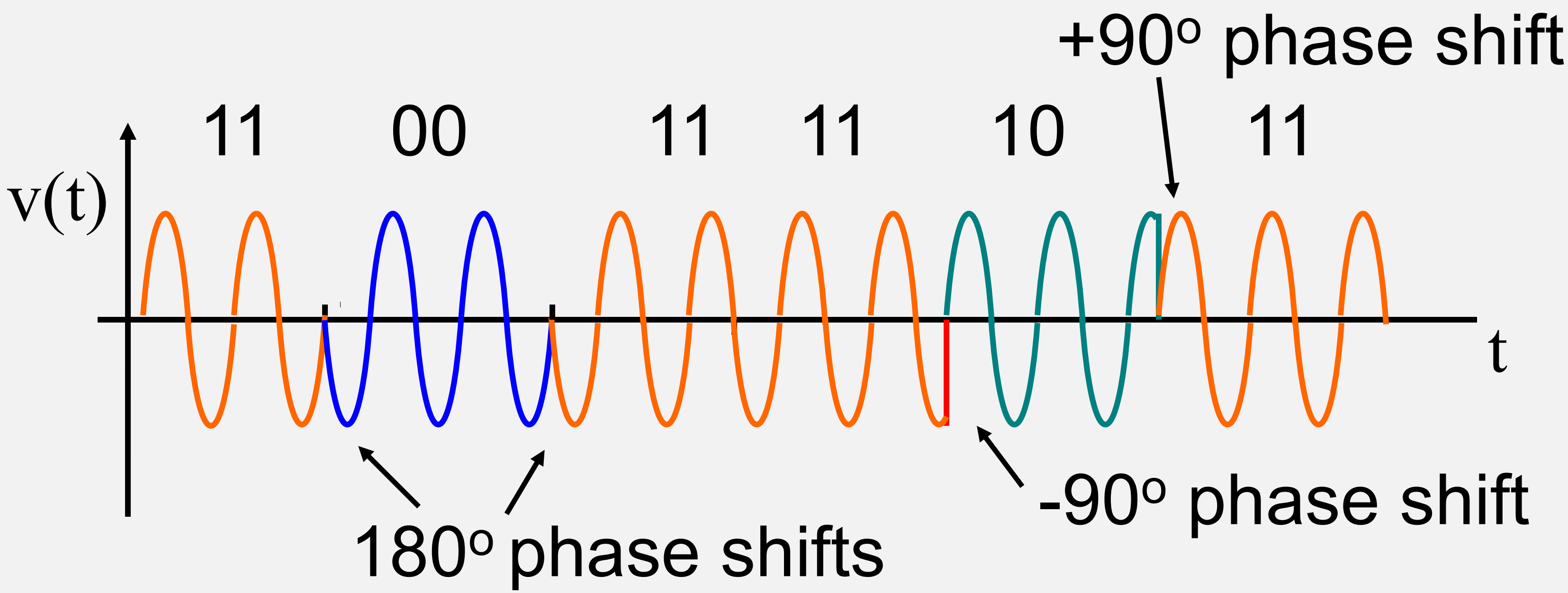
E.g., 00 is generated by $U_I(t) = -1$, $U_Q(t) = -1$

QPSK Waveform

- QPSK signal has form (unfiltered)

$$v_{QPSK}(t) = U_I(t)\cos\omega t + U_Q(t)\sin\omega t$$

$$U_I(t) \text{ and } U_Q(t) = +1 \text{ or } -1$$

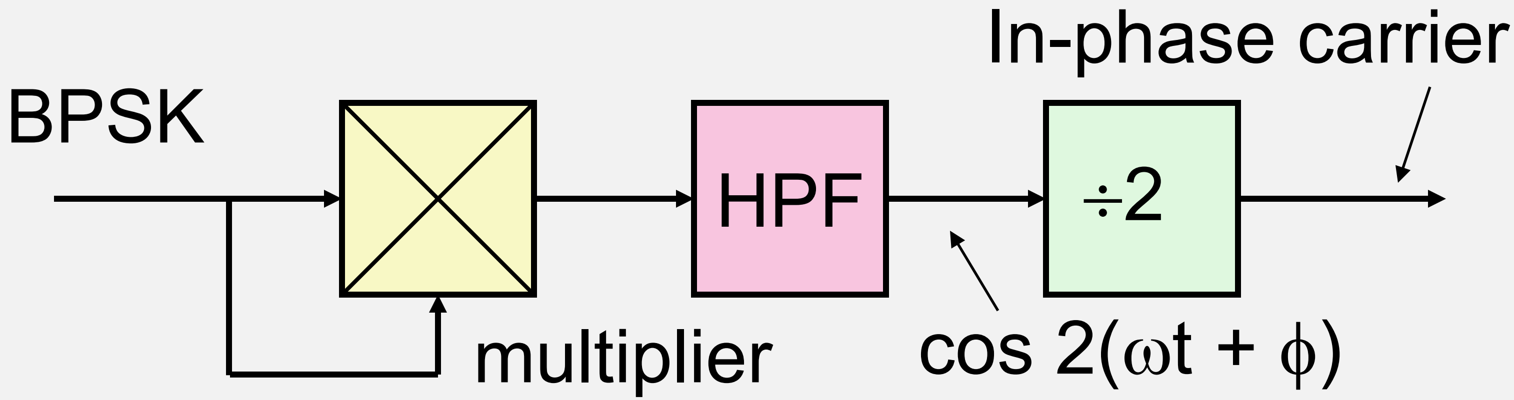


Carrier recovery

- Local oscillator in demodulator needs to be synchronized to received BPSK or QPSK carrier
- Must be derived from the received signal using a *carrier recovery circuit*
- For BPSK apply the received signal to both inputs of a mixer
- Obtain the square of the BPSK signal

Carrier recovery - BPSK

- Squaring circuit



$$v_o(t) = [\,U_I(t) \times V_{IF} \sin(\omega t\,)\,]^2$$
$$= U_I(t)^2 \times V_{IF}^2\,[\sin(\omega t)]^2$$
$$= 1 \times \tfrac{1}{2}\, V_{IF}^2\,[\,1 - \cos(2\omega t)\,]$$

HPF output is $\frac{1}{2}\, V_{IF}^2 \cos(2\omega t)$

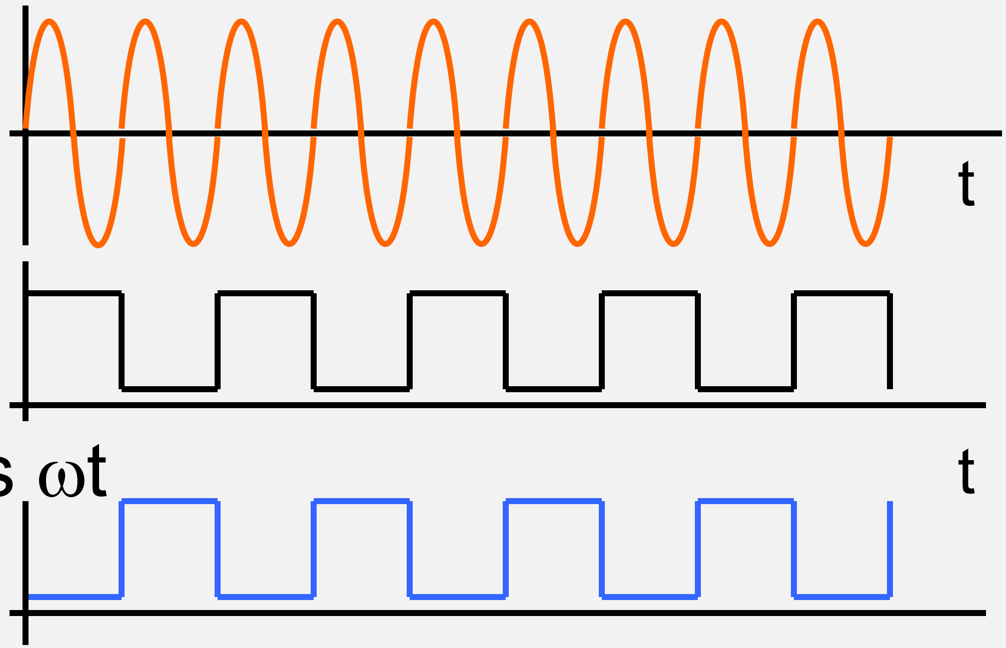
Carrier recovery

- For BPSK, square the signal and divide by two (using a PLL) to obtain in-phase carrier
- For QPSK, square the signal twice and divide by four to obtain in-phase carrier
- Dividing by two in BPSK carrier recovery circuit leads to ambiguity of 180^o
- Dividing by four in QPSK carrier recovery circuit leads to ambiguity of $n \times 90^o$, $n = 0$ to 3

Divide by two ambiguity

With BPSK ambiguity…

- cos $2\omega t$

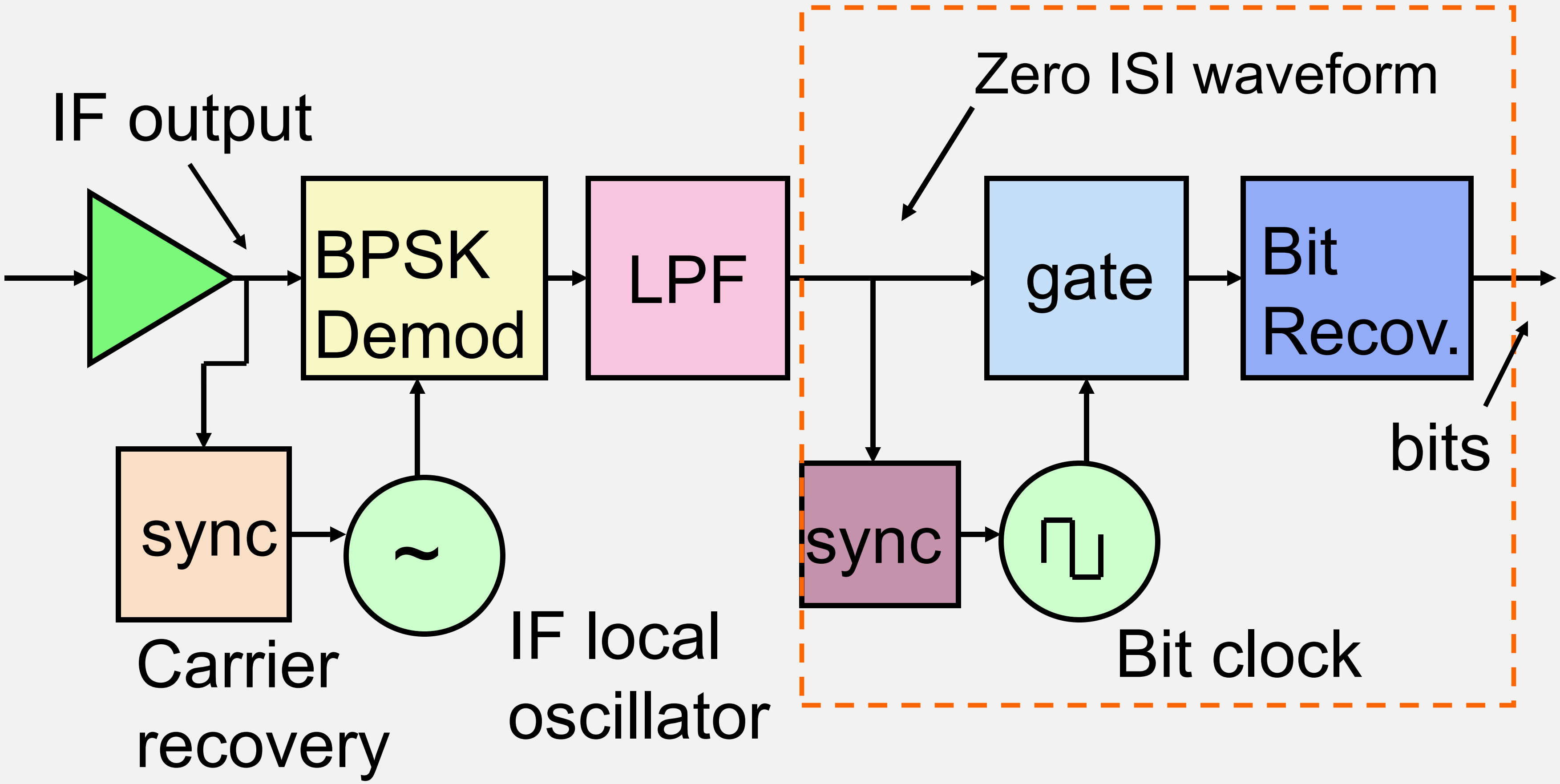


$\div 2$

$\Rightarrow$ cos ωt

…the complement of the bit stream could be produced (resolve with code words in bit stream)

Sampling and Bit Recovery



Summary (1/2)

- Digital transmission via satellite can use BPSK or QPSK
- Input to transmitter is NRZ rectangular pulses
- BPSK modulator is a multiplier
- BPSK demodulator multiplies by in-phase carrier
- Carrier and bit recovery is needed in receiver

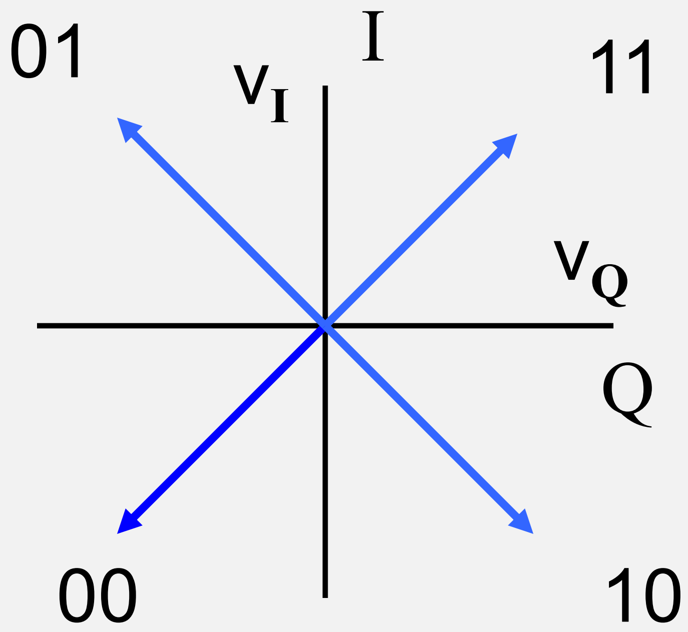
Summary (2/2)

- QPSK transmits two BPSK signals in phase quadrature
- QPSK signal carries two bits per symbol
- Can send same data rate as BPSK in half the bandwidth (but requires twice the power)
- QPSK demodulator is two BPSK demodulators in phase quadrature
- Carrier and bit recovery is needed in receiver

Gray Codes

- QPSK usually applies *Gray coding* to map pairs of data bits to phase angle
- Adjacent phase states then differ by only one bit (not two)
- ‘Regular’ noise causes only a one bit error
- Rate of occurrence of two bit errors is greatly reduced

QPSK Gray Coding



Gray coding: Each increment of phase changes only one bit, not two

Example of 4-bit Gray coding

Standard binary words map into Gray words

Reverse mapping on receive recovers the original binary words

Small noise fluctuations cause only 1 bit to be in error

Decimal	Binary	Gray
0	0000	0000
1	0001	0001
2	0010	0011
3	0011	0010
4	0100	0110
5	0101	0111
6	0110	0101
7	0111	0100
8	1000	1100
9	1001	1101
10	1010	1111
11	1011	1110
12	1100	1010
13	1101	1011
14	1110	1001
15	1111	1000

Gray coding: Summary

- Each phase state is associated with a particular 1- or 2-bit word
- With Gray coding the set of words is arranged such that the smallest increment of phase that can change a bit will not change more than one bit
- Advantage: Noise is less effective at causing multiple bit errors
- And 1-bit errors are easier to correct

Bit Error Rate (BER)

- We can calculate the probability of a symbol error from random noise theory
- We will always assume Additive White Gaussian Noise (AWGN)
- We need to know the relationship between C/N at the receiver input and Bit Error Rate (BER) at the receiver output

Bit encoding rules (binary)

- Recovery circuit in receiver outputs bits according to these rules:

At transmitter:

For data 1 send +V pulse

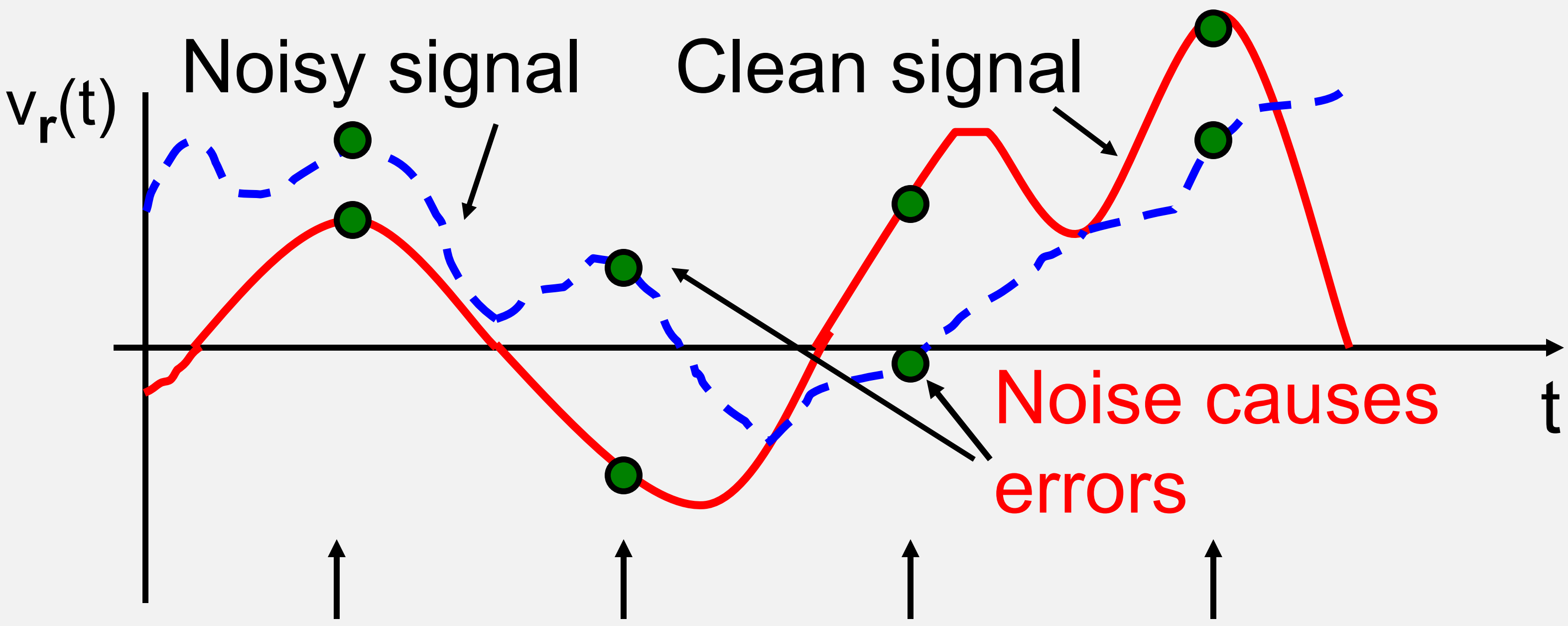
For data 0 send -V pulse

At receiver:

If $V_{\mathbf{sample}}$ **>** 0 volts, then data output = 1

If $V_{\mathbf{sample}}$ **<** 0 volts, then data output = 0

Sampling and bit recovery (binary)



Sample instants spaced T_o

Demodulator output waveform $v_r(t)$ is signal + noise

Symbol errors in binary receiver

- The output of a binary demodulator is

  $$v_{\mathbf{r}}(t) = v_{\mathbf{signal}}(t) + n(t)$$

  where n(t) is noise voltage at sample time

- If $v_{\mathbf{r}}(t) = v_{\mathbf{signal}}(t) + n(t) \mathbf{>} 0$ volts

  when a -V pulse was sent,

  we will output a data 1 $\Rightarrow$ a symbol error

Symbol errors in binary receiver

- If $v_{\mathbf{r}}(t) = v_{\mathbf{signal}}(t) + n(t) < 0$ volts

  when a +V pulse was sent,

  we will output a data 0 $\Rightarrow$ a symbol error

Symbol errors

- Condition for a symbol error to occur when a -V pulse is sent:

$$v_{\mathbf{signal}}(t) + n(t) > 0 \text{ volts}$$

i.e. $n(t) > |v_{\mathbf{signal}}(t)|$ or $n(t) > +V$

- Condition for a symbol error to occur when a +V pulse is sent:

$$v_{\mathbf{signal}}(t) + n(t) < 0 \text{ volts}$$

i.e. $n(t) < -|v_{\mathbf{signal}}(t)|$ or $n(t) < -V$

AWGN

- We will assume that the noise at the receiver output is AWGN …(even if it isn't)
- *AWGN = Additive White Gaussian Noise*
- Probability of a symbol error is

$$\tfrac{1}{2}\,P\,[\,(n(t) > +V \mid -V \text{ pulse sent}\,]$$

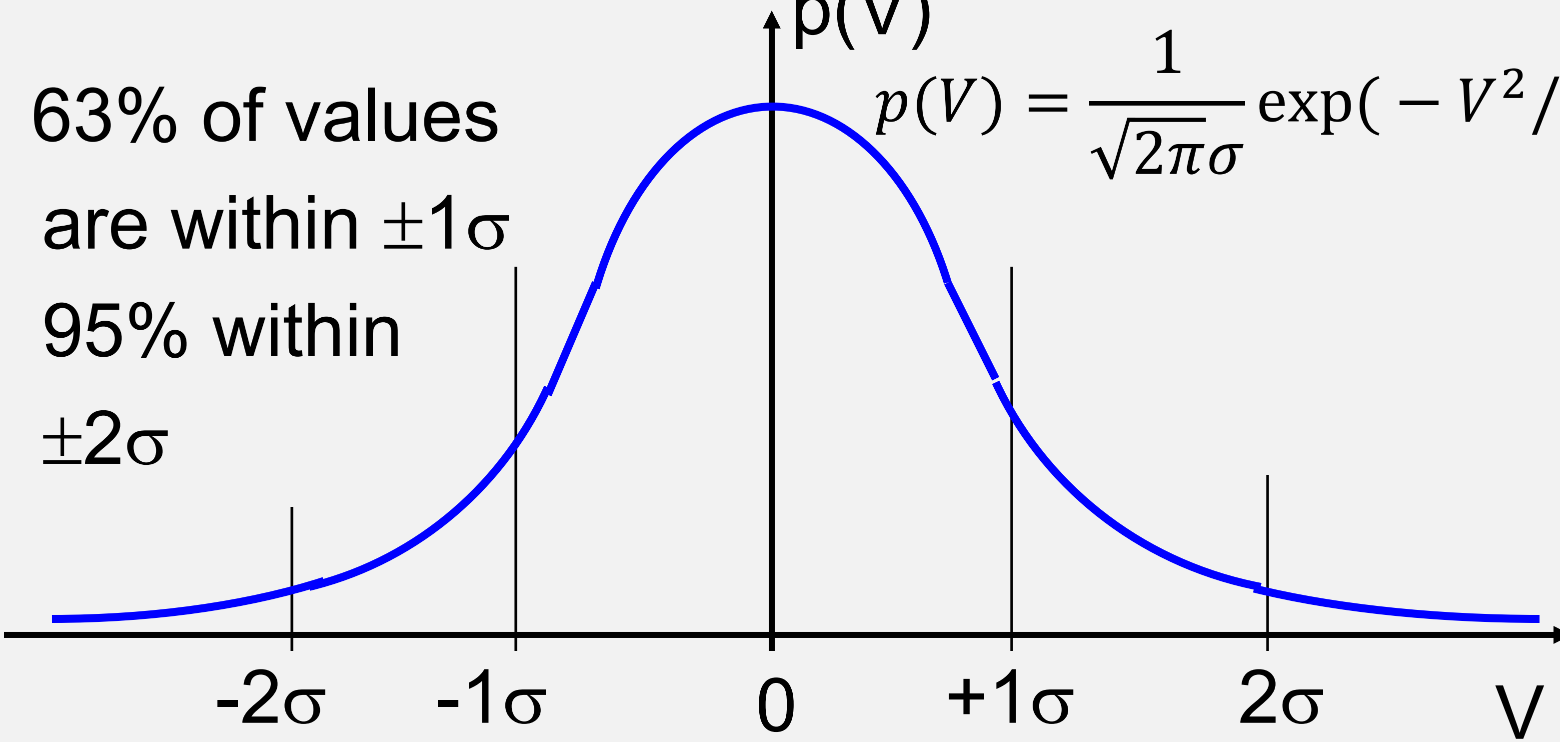
$$+\ \tfrac{1}{2}\,P\,[\,(n(t) < -V \mid +V \text{ pulse sent}\,]$$

$$=\ P\,[\,(n(t) > +V\,]$$

Gaussian Noise

- 63% of values are within ±1σ
  95% within ±2σ

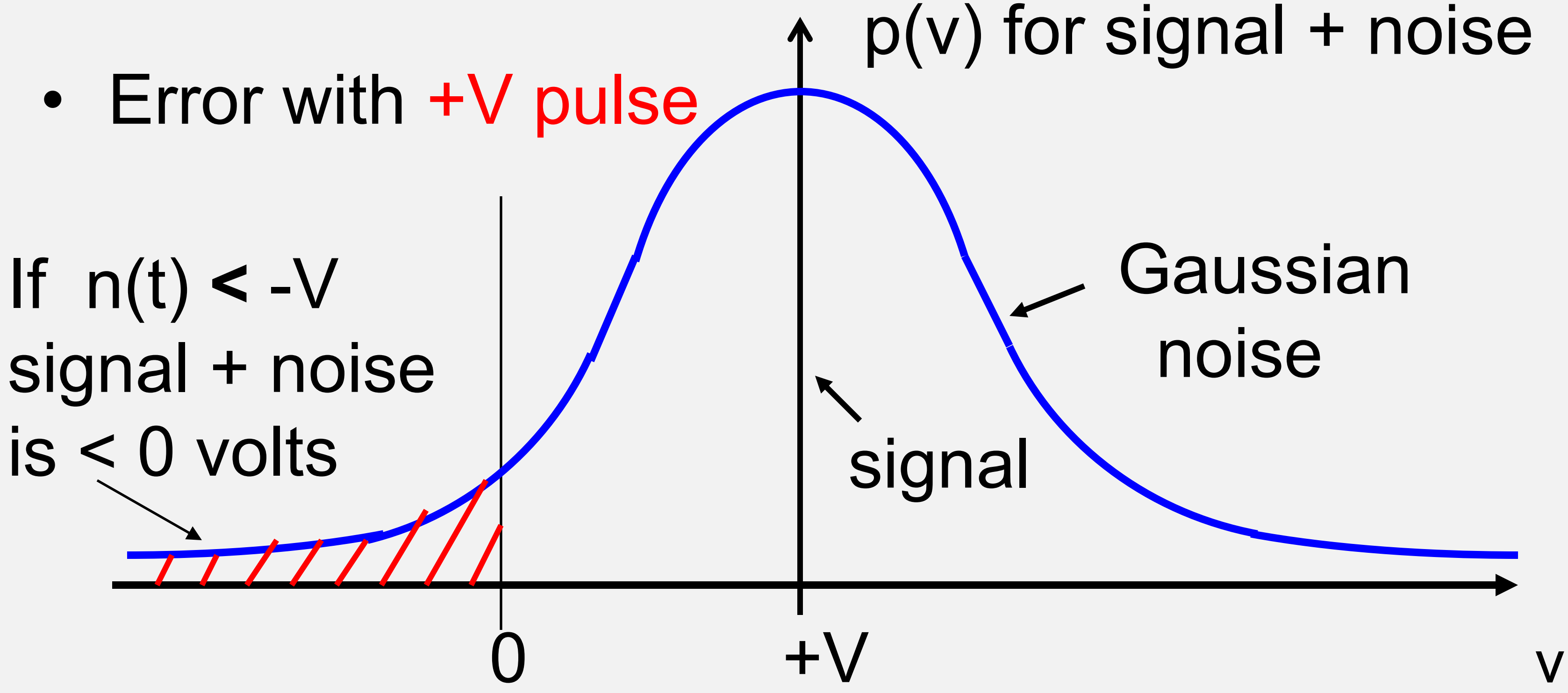
$$p(V) = \frac{1}{\sqrt{2\pi\sigma}} \exp(-V^2/2\sigma^2)$$



p(V) is a Probability Distribution Function (PDF)

Gaussian Noise Causes Errors

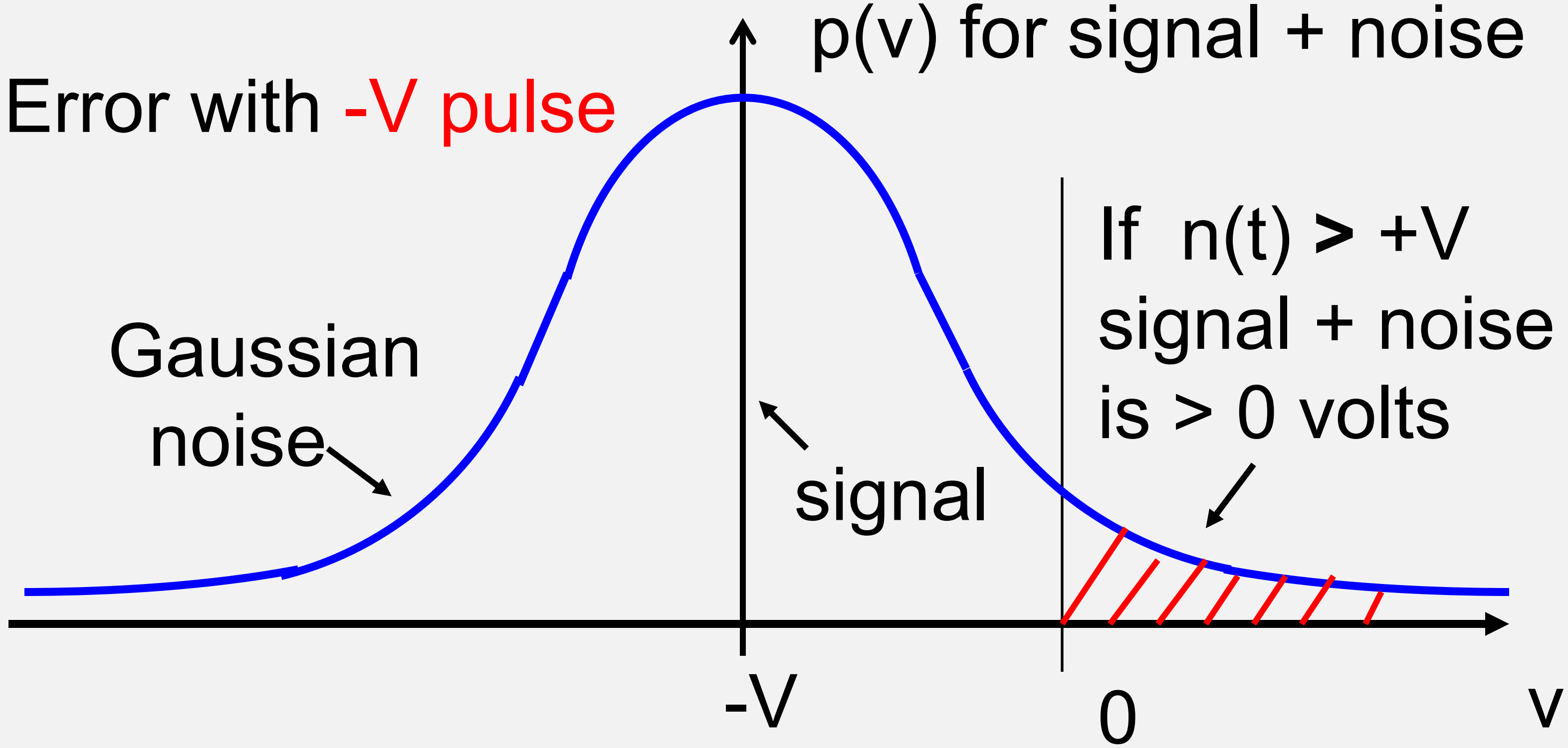
- Error with +V pulse



p(V) is a Probability Distribution Function (PDF)

Gaussian Noise Causes Errors

- Error with -V pulse



p(V) is a Probability Distribution Function (PDF)

Probability of Symbol Error

- Probability that we have a symbol error is

  $$P(\text{error}) = P\,[\,n(t) > +V\,]$$

  if we send equal numbers of V and –V pulses

- Probability of error is given by integral of p(v) from V to infinity

Probability of Symbol Error

- P(Error) is

$$\int_{+\mathrm{V}}^{\infty} p(V)\mathrm{d}V$$

- Where

$$p(V) = \frac{1}{\sqrt{2\pi}\sigma}\exp(-V^2/2\sigma^2)$$

- This integral (Eqn 5.21 in the text – note typos) has no analytical solution

- Solution is obtained in terms of the *Q function*, *Q(z)*, *complementary error function*, *erfc(x)*, or approximate expressions

Probability of Symbol Error

- For AWGN and an ideal binary link with RRC filters (derived in Section 5.3.1)

$$P(\text{symbol error}) = Q\left[\sqrt{(2E_b / N_o)}\right]$$

$$= \tfrac{1}{2}\,\text{erfc}\left[\sqrt{(E_b / N_o)}\right]$$

where E_b / N_o is the ratio of bit energy [Joules] to single-sided noise power spectral density [watts/Hz]

- With ideal RRC filters

$$E_b / N_o = C / N \qquad \text{[linear ratio]}$$

Probability of bit error

- We calculate C / N from the link budget assuming AWGN and an ideal link
- For a BPSK link:

$$P(\text{symbol error}) = Q\left[\sqrt{(2C / N)}\right]$$
$$= \tfrac{1}{2}\,\text{erfc}\left[\sqrt{(C / N)}\right]$$

- For a QPSK link:

$$P(\text{symbol error}) = Q\left[\sqrt{(C / N)}\right]$$
$$= \tfrac{1}{2}\,\text{erfc}\left[\sqrt{(C / 2N)}\right]$$

Probability of Bit Error

- Since we use Gray coding for QPSK phase states:

  Probability of 1-bit error $\approx$ Probability of symbol error

  Note: C/N values in these equations are always power ratios – ***NOT dB***

Bit Error Rate - BER

- Bit error rate is ***not a rate***
- BER is used in place of probability of a bit error
- Theoretical BERs derived in last few pages cannot be achieved
- There are no RRC filters, so some ISI occurs with real filters
- Timing jitter also causes ISI

Implementation margin

- To account for non-ideal link:
  - Subtract an *implementation margin* from C/N calculated from link budget
  - Do not include implementation margin in link budget - it is a parameter of the receiver
- Typical implementation margins
  - Low bit rates (up to T1): about 0.5 dB
  - High bit rates (up to OC 3): about 1 dB

Q (z) Function Table
See text - Appendix C

z	Q(z)
0	0.5
2.0	2.280 E-2
3.0	1.35 E-3
4.0	3.17 E-5
4.7	1.30 E-6
5.0	2.87 E-7
6.0	1.0 E-9

BPSK and QPSK Links

- BPSK link sends a two-state (Binary) symbol

$$R_b = R_s \qquad B_N = R_s = R_b$$

$$B_{occ} = R_s(1 + \alpha) = R_b(1 + \alpha)$$

- QPSK link sends a four-state (Quaternary) symbol

$$R_b = 2\,R_s \qquad B_N = R_s = ½\,R_b$$

$$B_{occ} = R_s(1 + \alpha) = ½\,R_b(1 + \alpha)$$

Useful BER numbers

- BPSK
  - Theoretical BER = 10^{-6} with C/N = 10.6 dB
- QPSK
  - Theoretical BER = 10^{-6} with C/N = 13.6 dB
  - Add implementation margin to theoretical C/N to obtain number for a link with real bandpass filters

BPSK Link Example

- BPSK link with α = 0.4 RRC filters
- Bit rate is 1.544 Mbps (T1)
- $(C/N)_o$ = 18 dB in clear air
- Implementation margin is 0.5 dB

a.) What is noise bandwidth of receiver?

b.) What is occupied bandwidth of RF signal?

c.) What is BER in clear air?

d.) What is BER if $(C/N)_o$ falls to 9 dB in rain?

BPSK Link Example

- For BPSK, symbol rate = bit rate = 1.544 Msps

  a.) Receiver noise bandwidth = R_s = 1.544 MHz

  b.) Occupied bandwidth

$$B_{occ} = R_s(1 + \alpha) = R_b(1 + \alpha)$$

$$= 1.544 \times 1.4 \text{ MHz}$$

$$= 2.16 \text{ MHz}$$

BPSK Link Example

c.) What is BER in clear air?

Clear air C/N = 18 dB

Implementation margin = 0.5 dB

$(C/N)_{\mathbf{effective}}$ = 17.5 dB $\Rightarrow$ 56.23

BER (= PE) = $Q\,[\sqrt{(2\,C\,/\,N)}]$

$= Q\,(10.60) \Rightarrow 0$

- If $z > 7.0$, there are effectively no errors within the lifetime of the link

BPSK Link Example

d.) What is BER if $(C/N)_o$ falls to 9 dB in rain?

Implementation margin = 0.5 dB

$(C/N)_{\mathbf{effective}}$ = 8.5 dB $\Rightarrow$ 7.08

From Q (z) table

BER (= PE) = Q [√ (2C / N)]

= Q (3.76) $\Rightarrow$ 8.6 x 10^{-5}

- The link will have R_b x BER ~ 100 errors per second (possibly tolerable)

QPSK Link Exercise

- QPSK link with α = 0.2 RRC filters
- Bit rate is 1.0 Mbps
- $(C/N)_{\mathbf{overall}}$ = 18 dB in clear air
- Implementation margin is 1.0 dB

a.) What is noise bandwidth of receiver?

b.) What is occupied bandwidth of RF signal?

c.) What is BER in clear air?

d.) What is BER if $(C/N)_{\mathbf{overall}}$ falls to 9 dB?

QPSK Link Example

- For QPSK, symbol rate = ½ bit rate = 0.5 Msps

a.) Receiver noise bandwidth = R_s = 0.5 MHz

b.) Occupied bandwidth

$$B_{occ} = R_s(1 + \alpha) = R_b(1 + \alpha)$$

$$= 0.5 \times 1.2 \text{ MHz}$$

$$= 0.6 \text{ MHz}$$

BPSK Link Example

c.) What is BER in clear air?

Clear air C/N = 18 dB

Implementation margin = 1.0 dB

$(C/N)_{\textbf{effective}}$ = 17.0 dB $\Rightarrow$ 50.1

BER (= PE) = Q [√ (C / N)]

= Q (7.08) $\Rightarrow$ 1.28 x 10^{-12}

- An error occurs on average once every nine days, i.e., practically never

BPSK Link Example

d.) What is BER if $(C/N)_o$ falls to 9 dB in rain?

Implementation margin = 1.0 dB

$(C/N)_{effective}$ = 8.0 dB $\Rightarrow$ 6.31

From Q (z) table

BER (= PE) = Q [$\sqrt{}$ (C / N)]

= Q (2.51) $\Rightarrow$ 1.2 x 10^{-2}

- The QPSK link in rain will experience so many errors (12,000 per sec) that it will probably shut down

Comparison of BPSK and QPSK

- With a fixed RF bandwidth, QPSK link sends twice as much data as BPSK link
- But error rate is much higher unless we can maintain a higher C/N
- Compared to BPSK link using same RF bandwidth, we need 3 dB more C/N overall at QPSK receiver input to get same BER

Comparison of BPSK and QPSK

- A QPSK link is just two BPSK links sharing the same RF channel
- We use phase quadrature to superimpose signals
- Each link gets half of transmitter power
- C/N for each BPSK link is 3 dB lower than case for single BPSK link
- Hence QPSK needs 3 dB more power to achieve same BER as BPSK

Multilevel Signaling

- We can have several bits per symbol
- QPSK has four states, two bits per symbol,
- 8-PSK has eight states, three bits per symbol
- m-PSK has m states, $m = 2^n$ n: # of bits per symbol
- Using *m-ary* modulations we can send more bits in a given bandwidth than binary can
- Penalty is that transmitter power goes up for a given BER

ASK

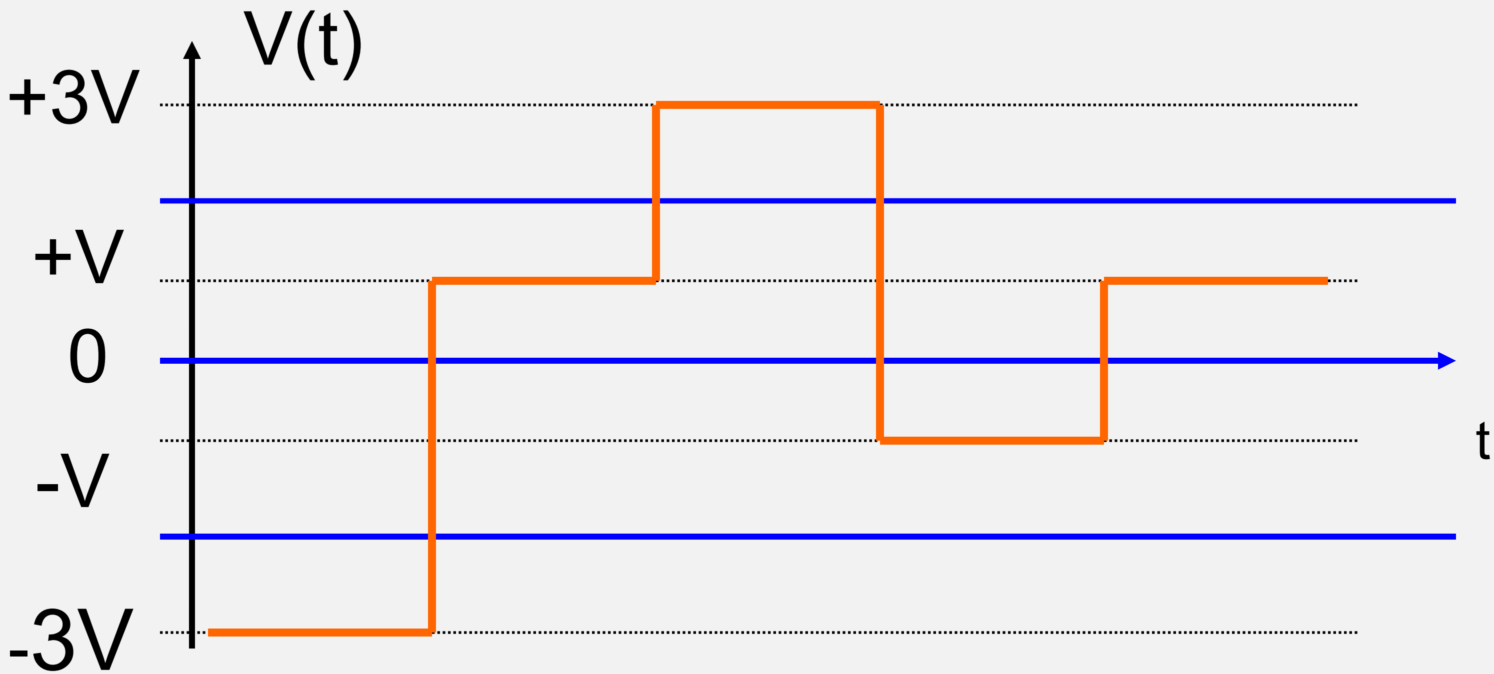
- Amplitude Shift Keying – modulate the carrier in amplitude
- Simplest case, binary-ASK
- One symbol, one bit, two possible states:
  - On / off
- 1 V / 3 V
- Rarely used

4-APSK Example

- APSK combines Amplitude and Phase Shift Keying
- 4-APSK sends two bits per symbol by combining binary-ASK with BPSK
- Voltage states are +3 V, +V, -V and -3V
- Probability of error is same as for binary-ASK with levels +V and +3V because separation of voltage levels is the same
- Penalty is more transmitter power is needed

4-APSK

Data 01 10 11 00 10



Thresholds are at +2V, 0V, and –2V (blue lines)

4-APSK Example

- 4-APSK encoding - use Gray coding to minimize occurrence of multi-bit errors

E.g. + 3V = 11

+ V = 10

- V = 00

- 3V = 01

4-APSK Power Example

- Power transmitted is the average power for a uniform distribution of symbols.
- Using one ohm as a reference for resistance, i.e., R = 1 ohm, we have $P_{\mathbf{av}} = \frac{1}{2} V_{avg}^{\mathbf{2}}$
- <u>For BPSK</u> with +1 V / -1 V :

$$P_{\mathbf{av}} = \tfrac{1}{2} \left[\tfrac{1}{2} \times V^2 + \tfrac{1}{2} \times V^2 \right]$$

$$= 0.5 \text{ W}$$

4-APSK Power Example

- For 4-APSK with +1 V / +3 V / -1 V / -3 V:

$$P_{av} = \tfrac{1}{2}\,[\tfrac{1}{4} \times 9 + \tfrac{1}{4} \times 1 + \tfrac{1}{4} \times 1 + \tfrac{1}{4} \times 9\,]$$
$$= \tfrac{1}{2} \times \tfrac{1}{4} \times 20$$
$$= 2.5\ \text{W}$$

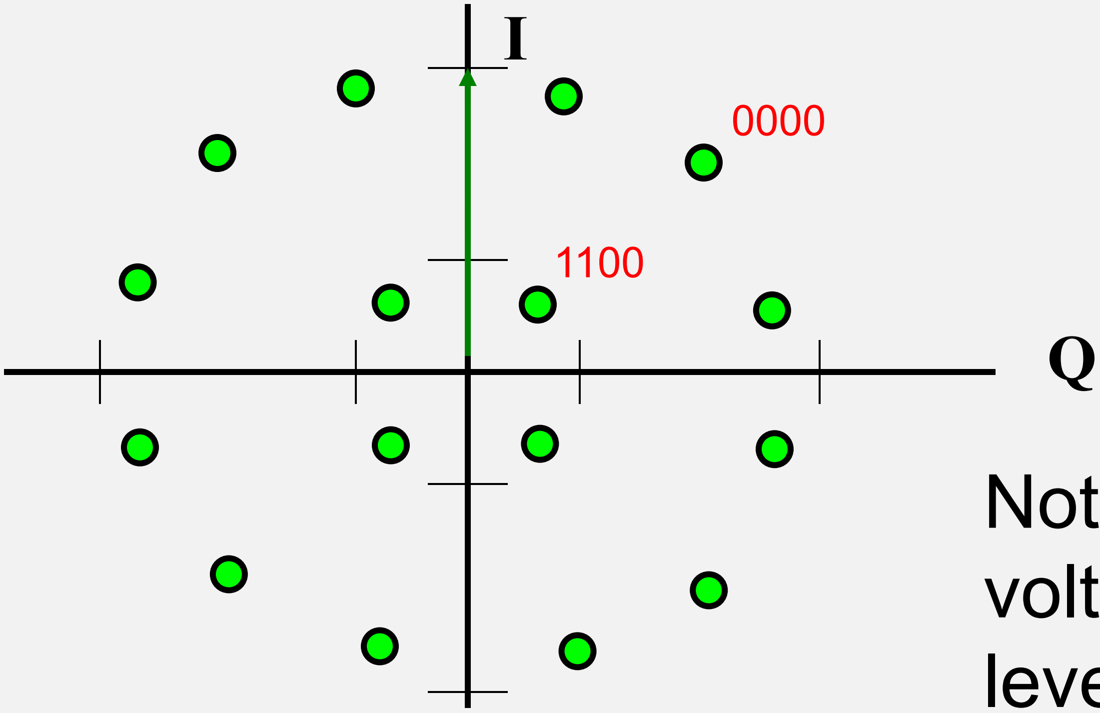
- This compares with 0.5 W for BPSK
- 4-APSK needs five times the power of BPSK

16-APSK Modulation

- 16-APSK increases the number of phase states to send 4 bits per symbol
- Still uses binary-ASK (two voltage levels)
- Higher power requirement for fixed error rate than 4-APSK

16-APSK Constellation Diagram

- Phasors are ready to have 4-bit words assigned with Gray coding



Note two voltage levels

QAM

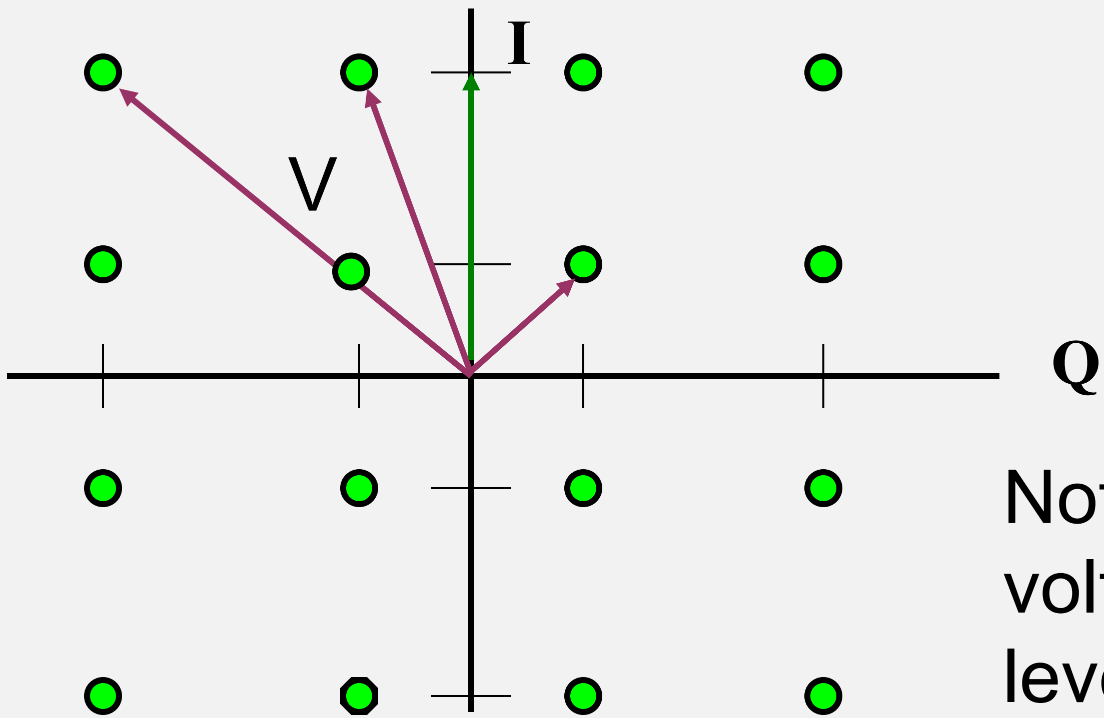
- Quadrature Amplitude Modulation: a combination of Q & A
- Q refers to use of two carrier channels in Quadrature (like QPSK)
- A refers to Amplitude modulation (multiple voltage levels)
- QAM is a combination of QPSK and m-ASK

m-QAM

- By combining phase states and voltage levels we can generate *m*-bit symbols
- Examples are 256-QAM (8 bits) and 1024-QAM (10 bits)
- 16-QAM sends 4 bits/symbol
- 16-QAM uses three voltage levels
- Recall that 16-APSK uses just two voltage levels

16-QAM Constellation Diagram

- Constellation diagram shows the tips of the I and Q phasors of the QAM signal



Note three voltage levels

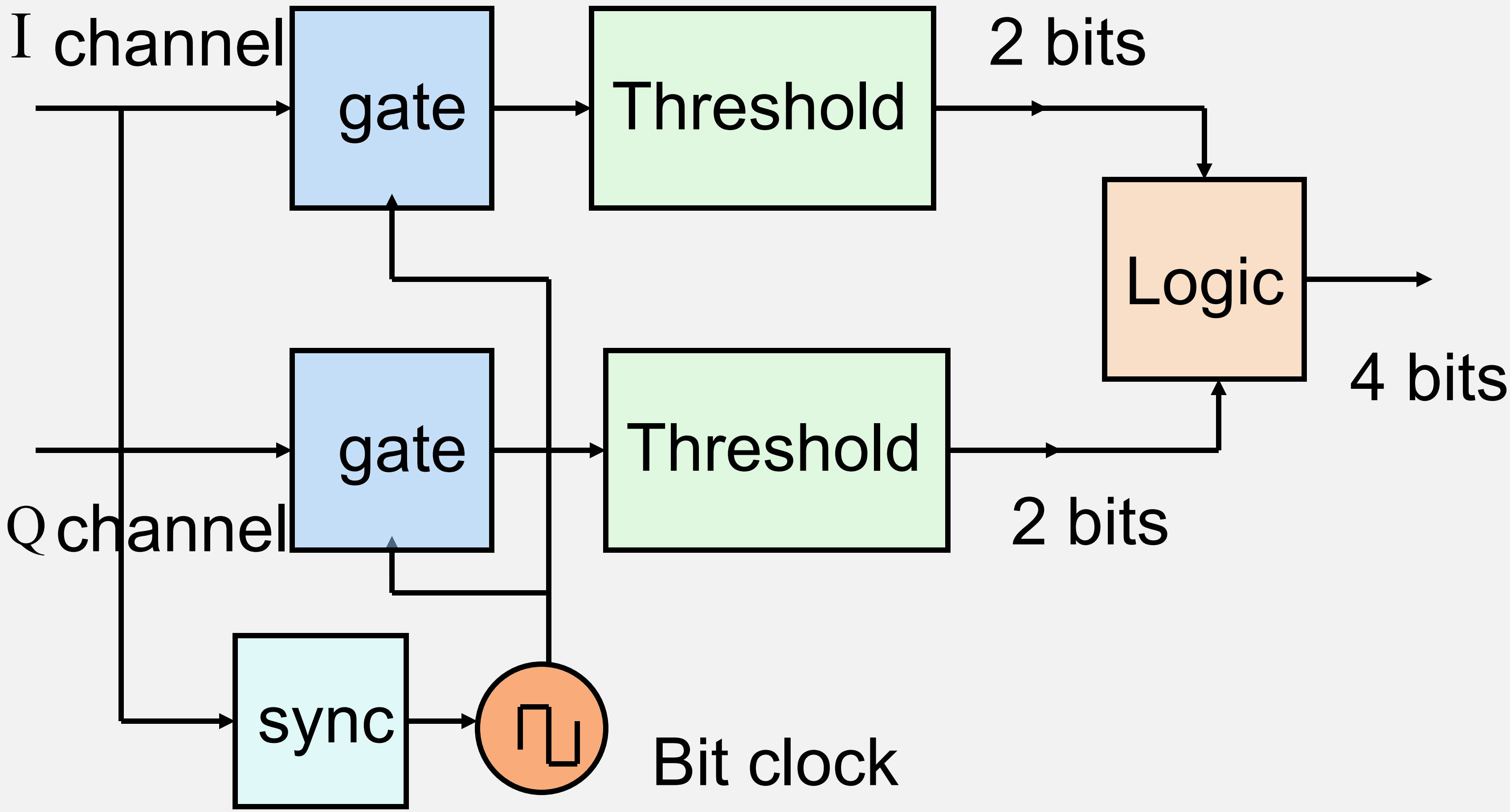
16-QAM modulator

- QAM operates with I and Q channels
- Logic at data input selects two bits for I channel and two bits for Q channel
- I channel has four possible states (2 x 2)
- Q channel has four possible states (2 x 2)
- There are 16 possible states of the symbol (2 x 2) x (2 x 2)

16-QAM demodulator

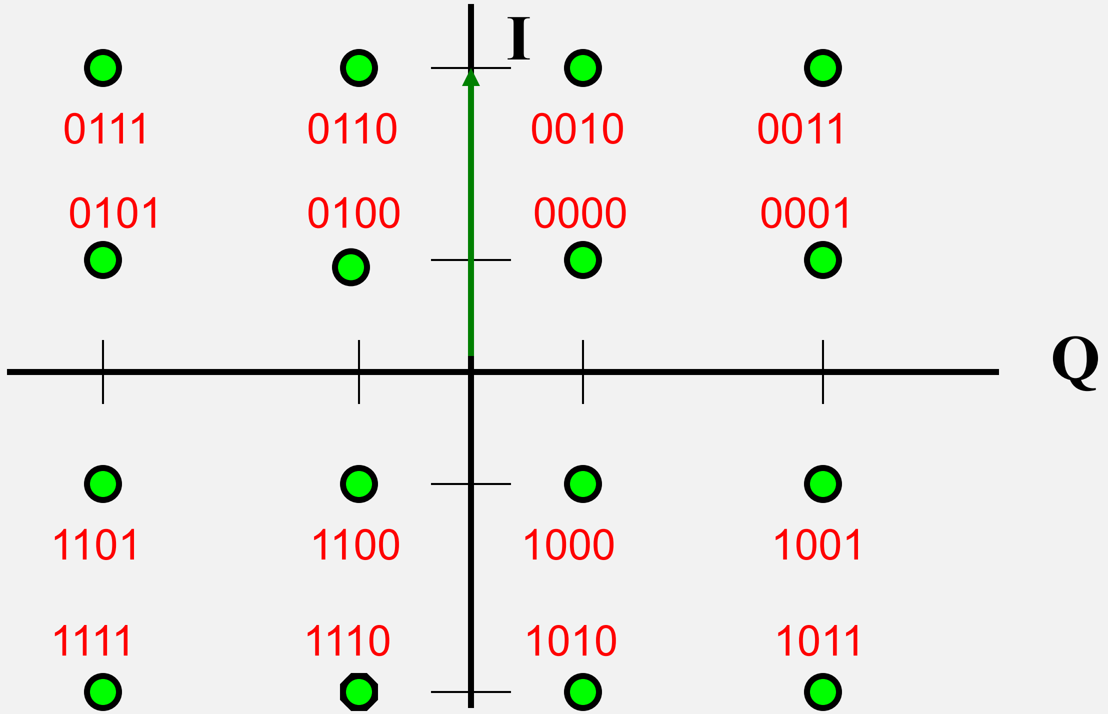
- Identical to QPSK demod except for logic block
- Need to add threshold detection for amplitude
- I channel generates two bits, Q channel generates two bits
- Logic block outputs four bits for each input symbol

16-QAM data recovery



16-QAM Constellation Diagram

- Constellation diagram with Gray-coded alphabet



16-QAM Constellation Diagram

- Observe that Gray-coding has been applied to have adjacent phasors differ by only one bit
- Three voltage levels: 4 phasors at lowest voltage, 8 at intermediate voltage, and 4 at highest voltage
- Phasors at the corners require considerably more power
- 16-APSK with two voltage levels might be a better choice for a non-linear transponder

C/N required for m-QAM / 16-APSK (BER = 10^{-6})

Moduln	Bits/symbol	C/N	Rel. B/W
BPSK	1	10.6 dB	1.0
QPSK	2	13.6 dB	0.5
16-QAM	4	20.5 dB	0.25
16-APSK	4	21.0 dB	0.25
32-QAM	5	24.4 dB	0.20
64-QAM	6	26.6 dB	0.17
256-QAM	8	32.5 dB	0.125
1024-QAM	10	38.5 dB	0.1

Artemis II Mission update - Optical Comm

https://www.scientificamerican.com/article/nasas-artemis-ii-laser-communications-system-is-beaming-4k-video-from-the/

- The main communications system for the Artemis II mission is RF (2.025-2.110 and 2.220-2.300 GHz)
- It is testing a laser-based system called Orion Artemis II Optical Communications System (O2O)
- O2O has been in use on the ISS
- Based on laser technology using bursts of signal at infrared frequencies
- Offers much higher bandwidth than RF
- The beam is very narrow, spanning only 6 km at Earth, requiring precise pointing
- Provides the basis for livestreaming clear images from the mission

Artemis II Mission – Earthset photo

https://www.scientificamerican.com/article/see-nasa-artemis-ii-missions-first-incredible-photos-of-the-moon-earth/

Artemis II Mission – Terminator photo

Forward Error Correction

- *Forward Error Correction* (FEC) can correct many of the errors at the receiver data output
- Penalty is a decreased data bit rate
- Add *parity* bits to the data stream
- Use parity bits to:
  – detect errors
  – correct errors
- Error detection is easier than error correction

Error Control

- Generically, error detection and error correction are known as *Error Control*
- FEC involves both detection and correction
- The ‘F’ in FEC implies that we do not go back to the source to correct errors

Error Control

- Error control is achieved with
  - Parity bits or cyclic redundancy checks
  - Block codes
  - Convolutional codes
  - Turbo codes
  - Interleaving

Block Code Example (Ch 5)

Message bits	Codeword
000	000 000
001	001 110
010	010 101
011	011 011
100	100 011
101	101 101
110	110 110
111	111 000

Block code example

- Simple example of a *systematic block code*
- Form is: message bits (k) + parity bits (n-k)
- Consists of 2^n possible codewords (n = number of bits in codeword)
- Here: n = 6 and 2^n = 64 codewords
- Number of valid message codewords: $2^3 = 8$
- Error detection / correction is simple - look for an invalid word and substitute nearest valid codeword (if unambiguous)

Detecting errors

Message bits	Codeword	Example Errors
000	000 000	010 000
001	001 110	101 110
010	010 101	011 100
011	011 011	100 011
100	100 011	101 100
101	101 101	110 010
110	110 110	101 001
111	111 000	000 111

Detecting and Correcting Errors

- In the Table in the previous slide, which of the entries:

  a. Has an error that can be *detected* ?

  b. Has an error that can be *corrected* ?

  c. Has an error that cannot be detected or corrected ?

Block code example

- Any codeword that is invalid as a message codeword is detected as an error
- Correction is possible if the error codeword is closer to one valid codeword than any other
- ‘Closer’ in the sense of requiring fewer bit flips to transform

Block code example

- Example: Received word is R = 1<u>0</u>0 110

  Nearest word is 110 110

- Differs by only one bit (easy to spot and fix)
- All other words in set differ from R by two or more bits
- The number of bits that are different between valid codewords creates *distance*

Error correcting block codes

- Example code has *minimum distance* = 3
- Minimum distance $d_{\mathbf{min}}$ is the smallest difference between any two valid codewords, i.e., the number of bits that have to be flipped to move from one codeword to the other
- Large minimum distance $\Rightarrow$ good correction properties
- Maximum number of bit errors sure to be <u>detected</u> = $d_{\mathbf{min}} - 1$

Error correcting block codes

- The ability to correct errors is lost when an invalid codeword can be mapped to more than one valid codeword (ambiguity issue)
- Maximum number of bit errors that can be <u>corrected</u> = ½ ($d_{\mathbf{min}}$ – 1)

  \- rounded down to next lower integer value
- An error cannot be detected when it creates another valid codeword!

Error correction block codes

- Example code has six bits
  - $2^6 = 64$ so there are 64 possible codewords
  - code set uses only 8, and $d_{min} = 3$
- We can correct 1-bit errors: $\frac{1}{2}(d_{min} - 1)$
- We can detect 2-bit errors: $(d_{min} - 1)$
- Some 3-bit errors cannot even be detected (x)

Detecting errors

Message bits	Codeword	Example Errors	
000	000 000	010 000	c
001	001 110	101 110	c
010	010 101	011 100	d
011	011 011	100 011	x
100	100 011	101 100	
101	101 101	110 010	
110	110 110	101 001	
111	111 000	000 111	

Error correcting block codes

- To correct 2-bit errors we need $d_{\mathbf{min}} = 5$
- BCH, Golay, and Reed-Solomon codes are examples of Block Codes
- Some BCH block codes can correct 10-bit errors
  - Music CDs all use Reed-Solomon coding
  - DBS-TV uses Reed-Solomon coding

Error correcting block codes

- Block codes have form (n, k)
  - n = number of bits in codeword
  - k = number of message bits
  - (n – k) = number of parity bits
- Code rate = k / n
- If k = ½ n we have a *half rate* code
- If k = ¾ n we have a *¾ rate* code

Golay codes

- Golay block codes are, in a sense, *perfect*
- All possible patterns of a given number of errors are corrected
- The (24,12) Golay code detects all patterns of 7-bit errors and corrects all patterns of 3-bit errors
- Note that it has a coding rate of ½, which is easier to implement in terms of timing

Convolutional coding

- Better code performance can be obtained with *convolutional* coding and decoding
- Convolutional encoder generates a new, longer bit stream - message bits are no longer visible
- Each input bit is spread over several adjacent output bits
- *Constraint length* is number of output bits influenced by one input bit

Convolutional Encoder

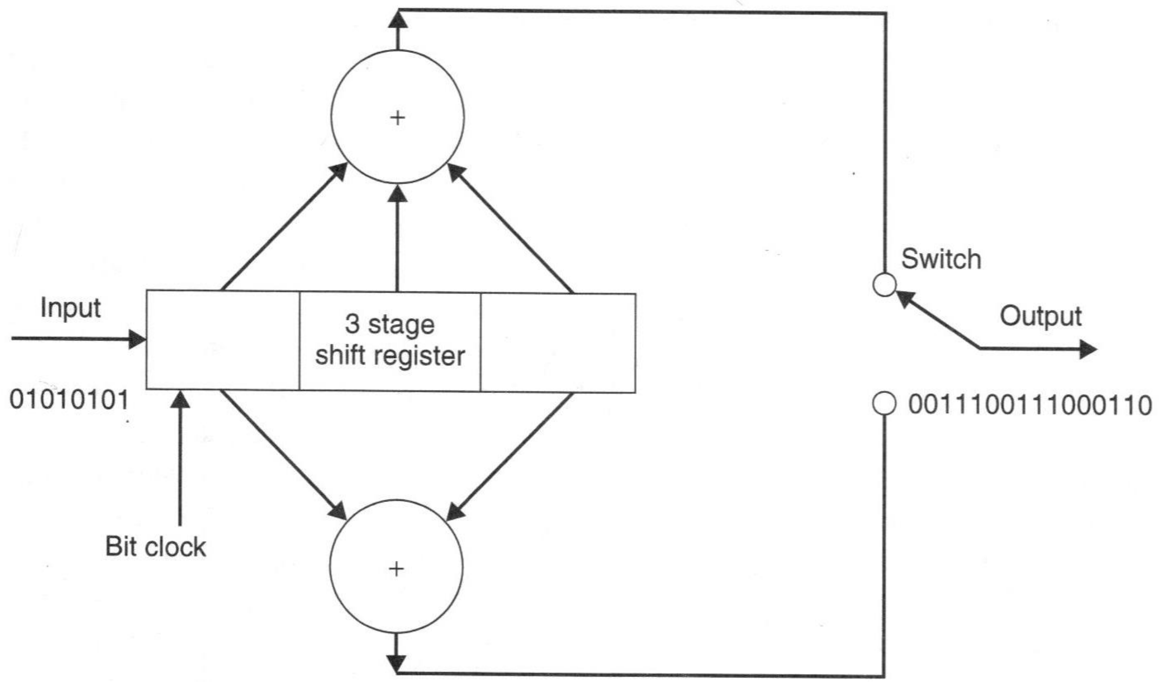


FIGURE 7.5 A simple convolutional encoder. The switch changes state at each input bit clock transition.

Pratt, Bostian, Allnut, 2nd ed.

Convolutional coding

- Spreading of information between input and output bit streams is achieved with a shift register (a sequence of bits)
- Invalid transitions are detected on receive and corrected
- Longer constraint length = better error correction properties

Convolutional coding

- Convolutional coding requires *maximal likelihood* decoder
- The decoder works through all possible tracks leading to a recorded error and selects the most probable
- Viterbi algorithm is one example
- Constraint length 7 and 8 codec chips are available as ICs

Coding gain

- Coding gain makes it possible to operate a link with a lower $(C/N)_o$ for a given BER
- Coding gain

  $= (C/N)_{o\ min}$ w/o coding - $(C/N)_{o\ min}$ w/ coding
- E.g. coding gain is an *effective increase* in C/N for a given BER

Coding gain

- Example: BER = 4×10^{-3} from QPSK receiver with

  $(C/N)_o$ = 9.0 dB and Imp. margin = 0.5 dB

- Coding IC with ½ coding rate has published coding gain of 6 dB, so $(C/N)_o$ = 15.0 dB

- BER out of decoder is then equal to BER for

  $(C/N)_o = 9.0 - 0.5 + 6.0 = 14.5$ dB

- From Q(z) table, BER = $Q(5.31) = 6 \times 10^{-8}$

Coding gain

- BER out of decoder is 6×10^{-8}, was 4×10^{-3}
- A big improvement **but** published coding gain requires ½ coding rate..

  ..so the message bit rate has fallen by one half!
- To restore the message bit rate, we must double the bit rate, symbol rate, and transmission bandwidth
- Then noise power increases by 3 dB and $(C/N)_o$ falls by 3 dB

Coding gain

- Noise power increases by 3 dB, $(C/N)_o$ falls by 3 dB
- New $(C/N)_o$ is actually $15.0 - 3 = 12.0$ dB
- If we want to send the same data rate

  $(C/N)_{\mathbf{effective}} = 12.0 - 0.5 = 11.5$ dB

  New BER $= Q(\sqrt{14.12}) = Q(3.76)$

  $= 8.6 \times 10^{-5}$ (versus 4×10^{-3})
- Still an improvement but it is less than the advertised 6 dB!

Coding gain

- Implementing FEC and coding gain entails a reduced rate of data transfer
- Hence terms such as *half-rate* FEC
- FEC is often implemented as ICs which can be switched in or out
- The price for coding gain:
  - a drop in the data transfer rate, or
  - an increase in bandwidth and hence noise

Burst errors

- The error correcting codes do well when bit errors occur singly
- But bit errors tend to occur in bunches called *bursts*
- Most error correcting codes have limited *burst error correction* capability

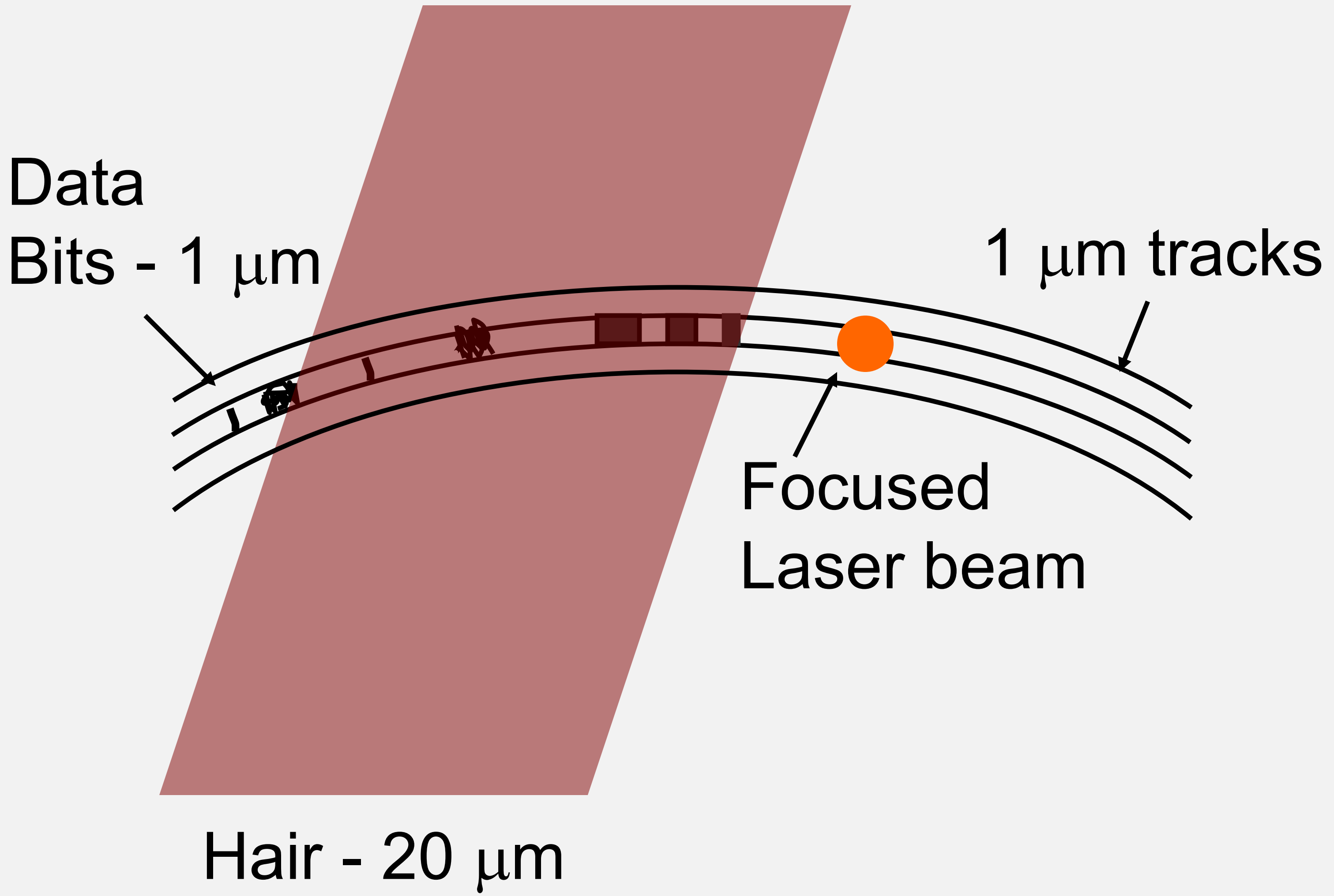
Interleaving

- *Interleaving* spreads out the errors in a burst and makes correction easier
- Often used with *concatenated* coding
  - two layers of coding … inner and outer
- Interleaving doesn't correct bit errors, it isolates them (which makes them easier to correct)

Interleaving

- Burst error occurs when a sequence of bits is corrupted
- Interleaving is used to spread out burst errors
- Example: Hair on a CD recording
- Bit length = 1 micron (= 1 μm)
- Scratch width = 20 microns
- Bits corrupted = 20

Compact Disk - CD



Interleaving

- *Interleaver* at the transmit (record) end reads data into a table by row
- Receiver (replay) reads out data by columns
- Bit errors caused by bursts are spread out
- Used on CD and DVD recordings, digital TV via satellite (DBS-TV)

Interleaver

Read in

1	6	11	16	21
2	7	12	17	22
3	8	13	18	23
4	9	14	19	24
5	10	15	20	25

Read out

Boxes show bit numbers in data sequence

Interleaving

- 5 x 5 interleaver example
- Read in sequence

  1 2 3 4 5 6 7 8 9 0 11 12 13 14 15 16 17 …
- Read out sequence

  1 6 11 16 21 2 7 12 17 22 3 9 13 19 23 …
- Example of burst error corrupting adjacent bits

  1 6 11 16 21 * * * * 22 3 9 13 19 23 …

De-interleaving

- Corrupted sequence is de-interleaved at receiver
- Received sequence

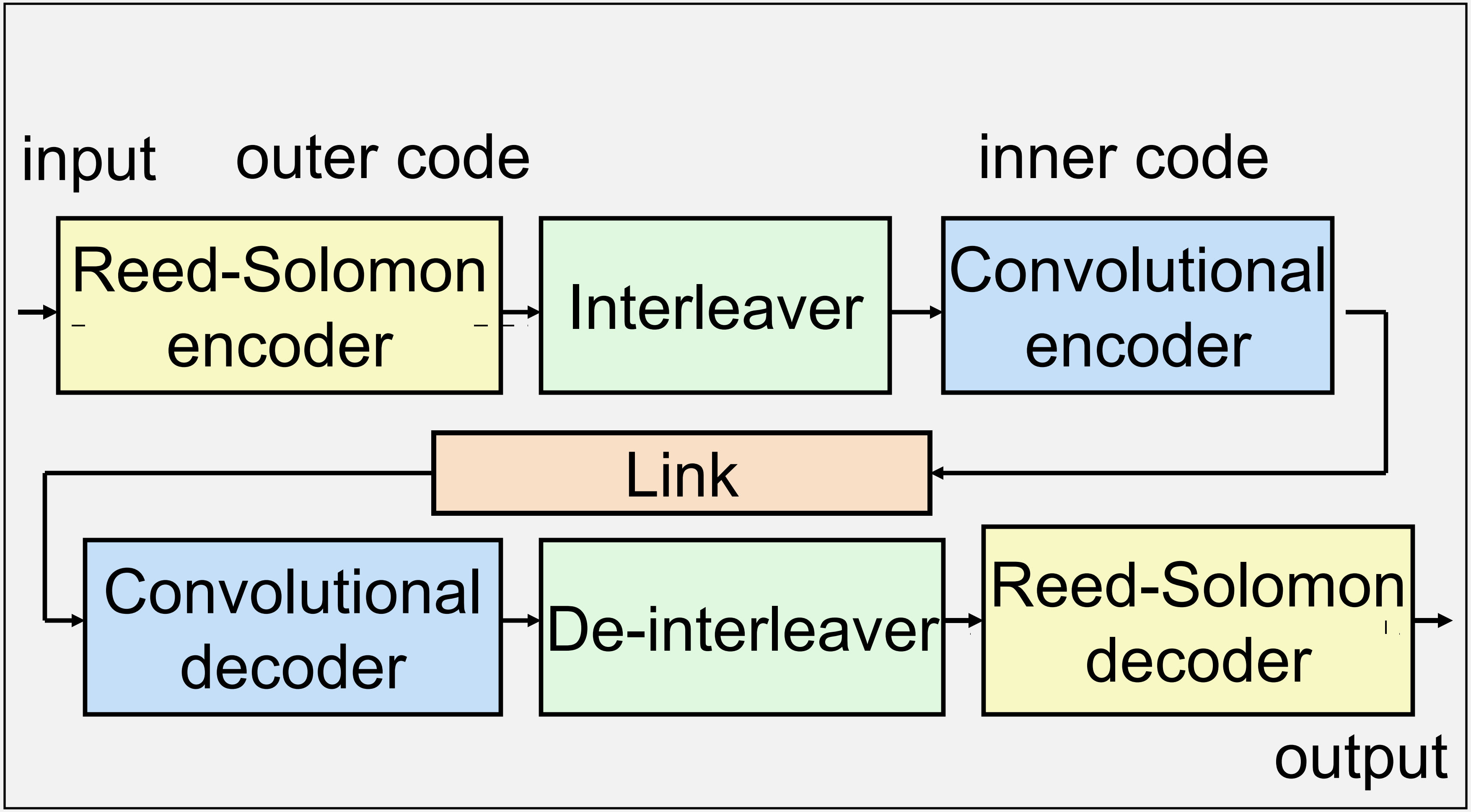
  1 6 11 16 21 * * * * 22 3 9 13 19 23 …
- After de-interleaver

  1 * 3 4 5 6 * 8 9 0 11 * 13 14 15 16 * …
- Bit errors from the burst are spread out and more correctable

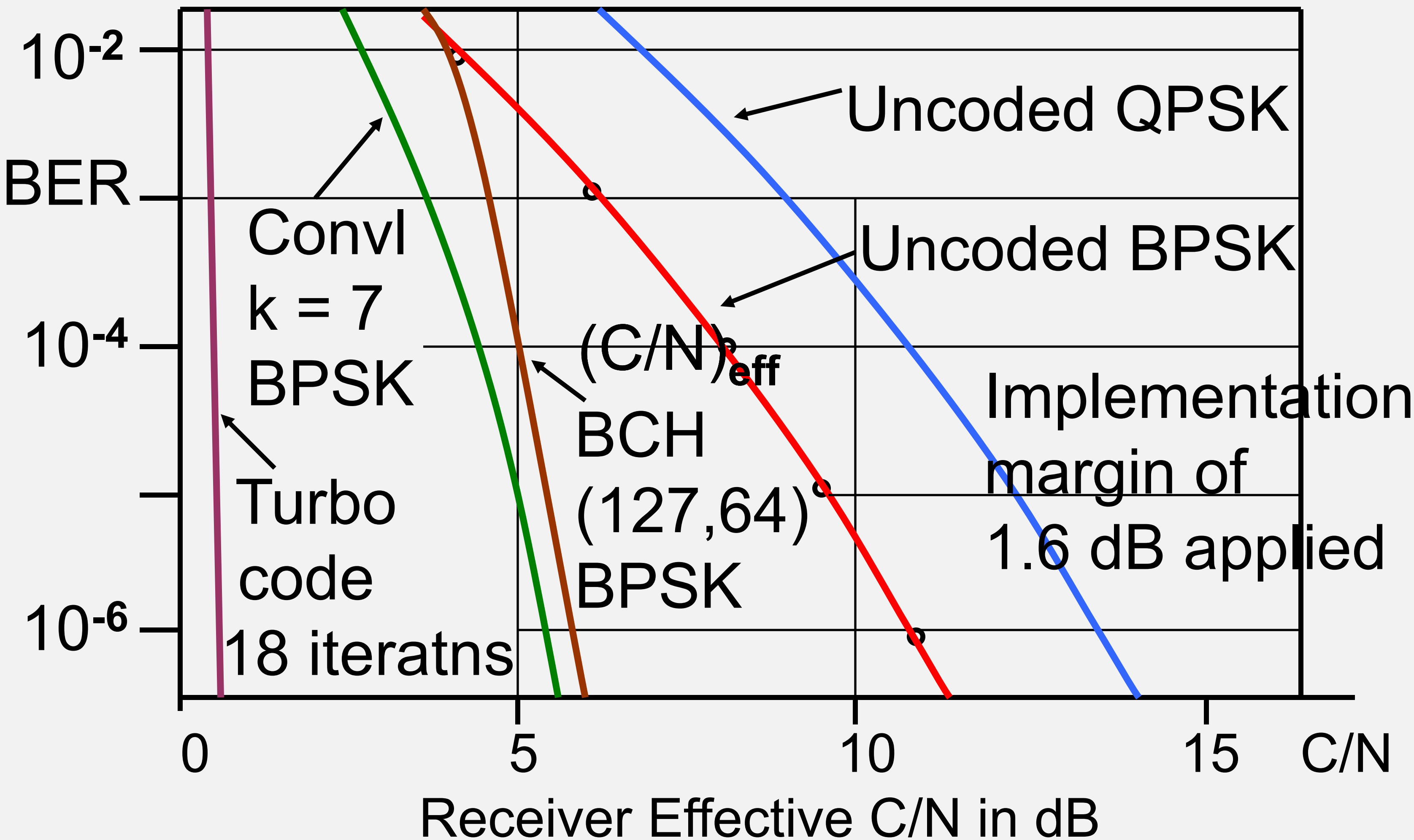
Forward Error Correction

- Example: Direct Broadcast Satellite TV
- Double layer of coding with interleaving:
  1) Outer layer = Reed-Solomon 204 / 188
  2) Interleaver = 17 x 11 typical
  3) Inner layer = Convolutional, k = 7
- Coding gain = 5.5 dB at output BER = 10^{-6}

FEC sequence



BER versus FEC technique (DVB-S)



Summary

- BER is related to (C/N) and modulation type
- Need Implementation Margin (IM) to find BER
- QPSK sends twice as many bits as BPSK in same RF bandwidth, needs twice the C/N
- *m-ary* modulations require higher C/N as *m* increases to maintain BER
- Error correction improves BER and leads to coding gain

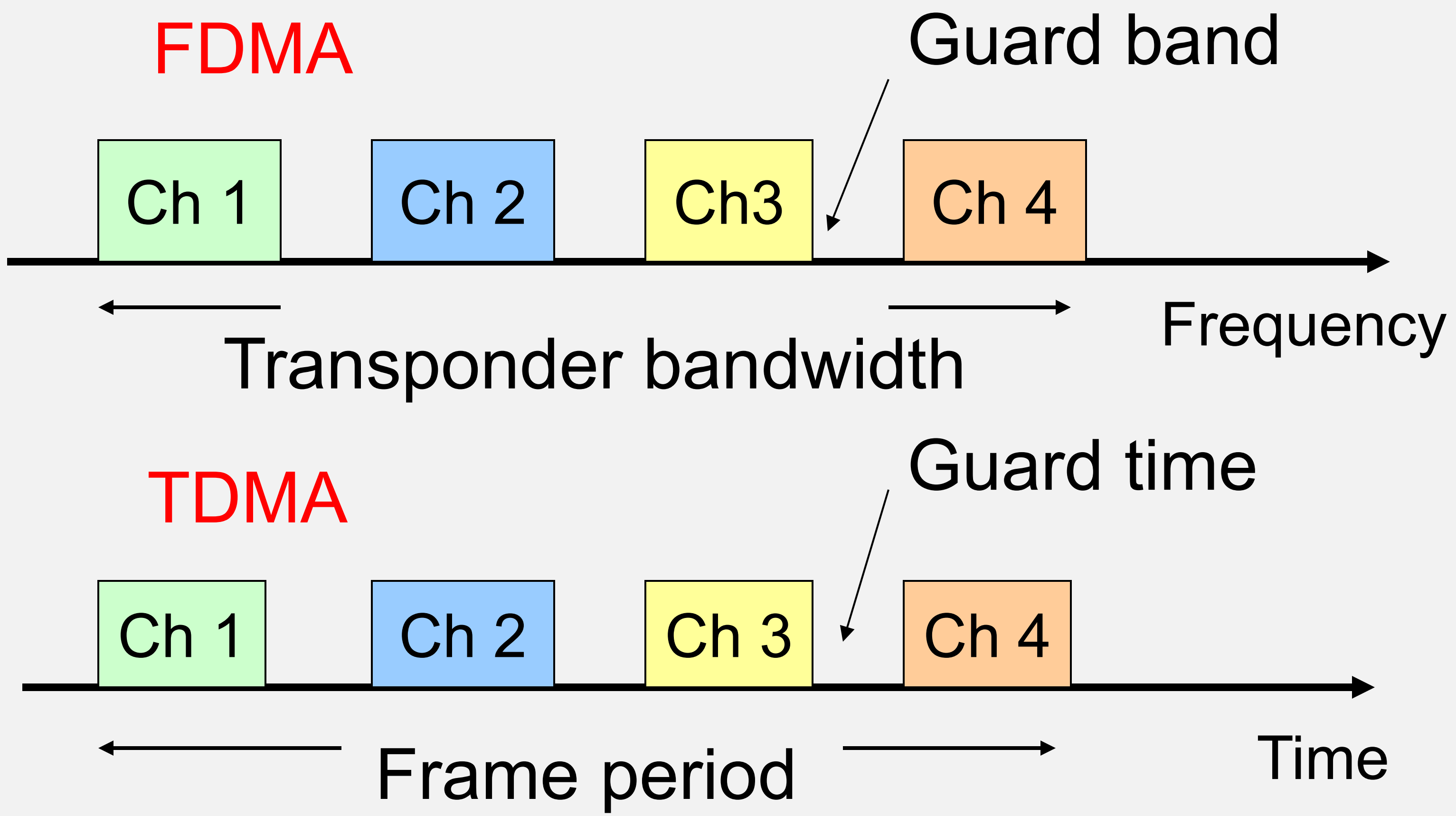
Multiple Access Techniques

- Communication satellites carry many signals at the same time using *Multiple Access* (MA) techniques
- These make use of frequency, time, or code to label individual signals in the presence of other signals
- The choice of MA technique is a design consideration for a Satcom system

MA Techniques

- FDMA (Frequency Division Multiple Access)
  - Signals are separated in frequency
- TDMA (Time Division Multiple Access)
  - Signals are separated in time
- CDMA (Code Division Multiple Access)
  - Signals are separated by coding
- Random Access (RA)
  - Signals are transmitted independently of one another

Fig. 6.9


FDMA
Guard band
Ch 1
Ch 2
Ch3
Ch 4
Frequency
Transponder bandwidth
TDMA
Guard time
Ch 1
Ch 2
Ch 3
Ch 4
Time
Frame period

FDMA / TDMA / CDMA / RA

- With FDMA, signals are assigned to distinct bands within a designated transponder bandwidth
- With TDMA, signals occupy distinct time slots
- With CDMA, transmission occurs at the same frequency and time; coding is used to extract individual signals
- With RA, there is no effort to separate the signals

FDMA - Tx

- FDMA is widely used as a way to share transponder bandwidth using multiple carrier frequencies and guard bands
- Transponder bandwidths of 36, 54, and 72 MHz are typical
- Earth stations are assigned to channels within the bandwidth

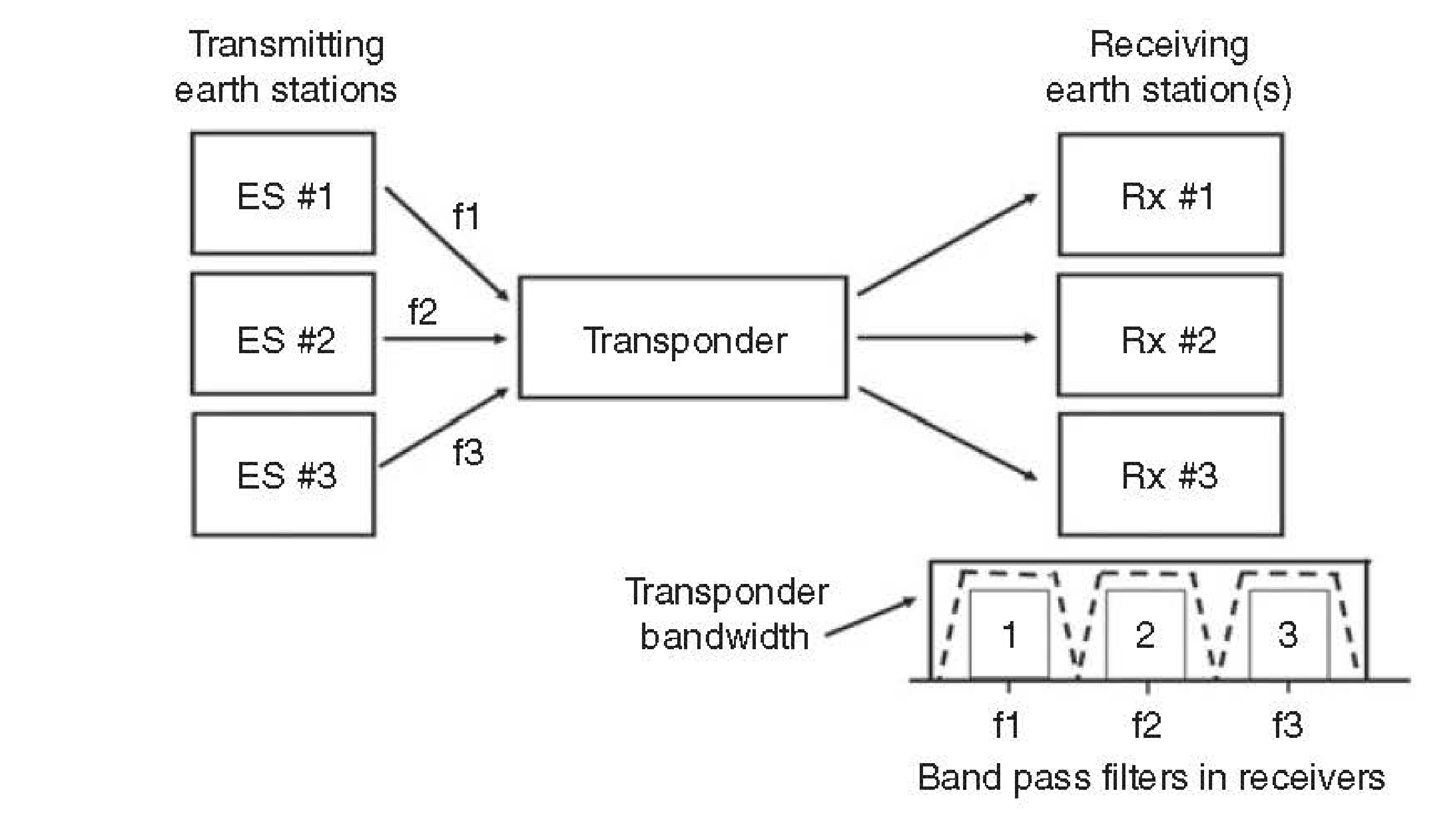
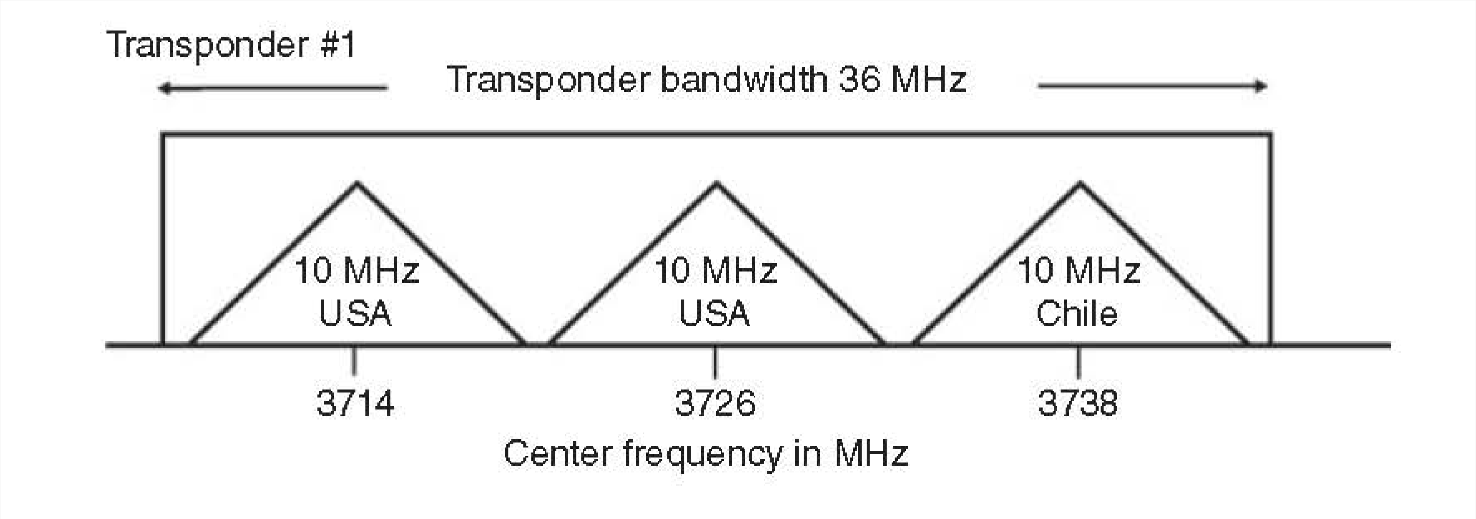


Illustration of FDMA (Figure 6.11 from course text)

FDMA - Tx

- For Tx the signals are mapped at the IF stage to distinct frequency bands
- These upconvert to a mapping of frequency bands about the RF carrier frequency
- The frequency bands fit within the transponder bandwidth with guard bands applied



FDMA plan for a C-band transponder (from Figure 6.12 from course text)

FDMA - Rx

- For Rx the Earth station downconverts all the signals to IF
- Selects for the desired signal frequency with a narrowband filter
- A large numbers of IF receivers is needed if the station wants to be able to receive all the signals
- Observe that the noise situation for FDMA is determined by the narrowband Rx filter

FDMA – Intermodulation Products

- Interference between signals in the transponder gives rise to Intermodulation products (IM), a type of noise that can be associated with a $(C/N)_{IM}$
- Encountered when transponder operates near saturation (high power), in the nonlinear regime
- IM is countered by applying backoff to the transponder power, which lowers $(C/N)_{dn}$
- Systems are designed to find the 'sweet spot' in terms of trading off $(C/N)_{dn}$ and $(C/N)_{IM}$ to maximize $(C/N)_o$

FDM versus FDMA

- Do not confuse *Frequency Division Multiplexing* (FDM) with FDMA
- FDM refers to signals being multiplexed onto a transmission link at a single Earth source
- FDMA has multiple Earth sources of signal
- FDM can be combined with FDMA to produce *FDM – FDMA*
- Many such hybrid modes are possible

TDMA

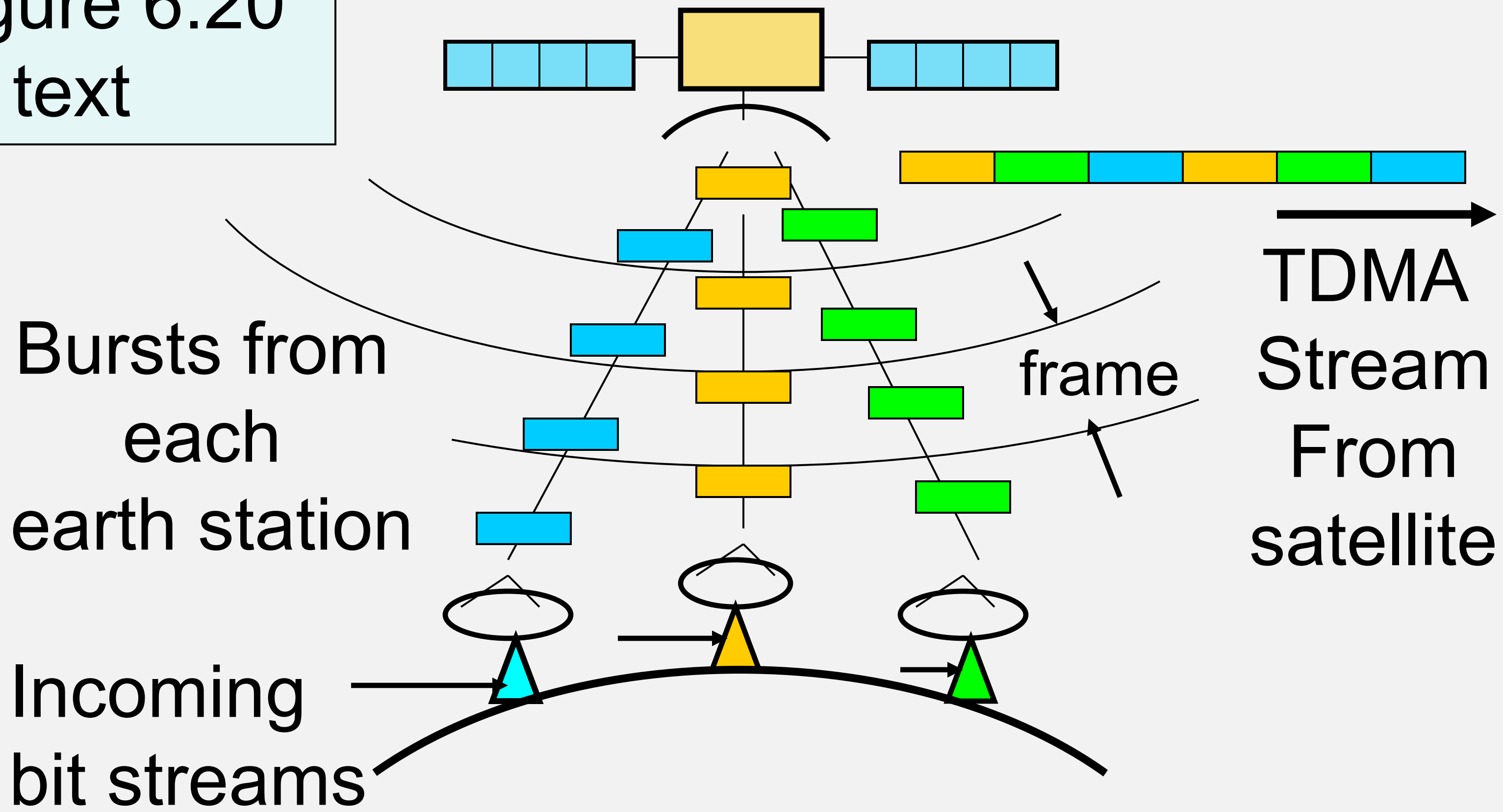
- Signals are separated in time
  - Transponder bandwidth is common to all signals
  - Stations transmit *bursts* of signal in sequence
- Each earth station transmits *in sequence*
- Transmitted bursts from many earth stations arrive at the satellite

    - in an orderly fashion

    - in the correct order

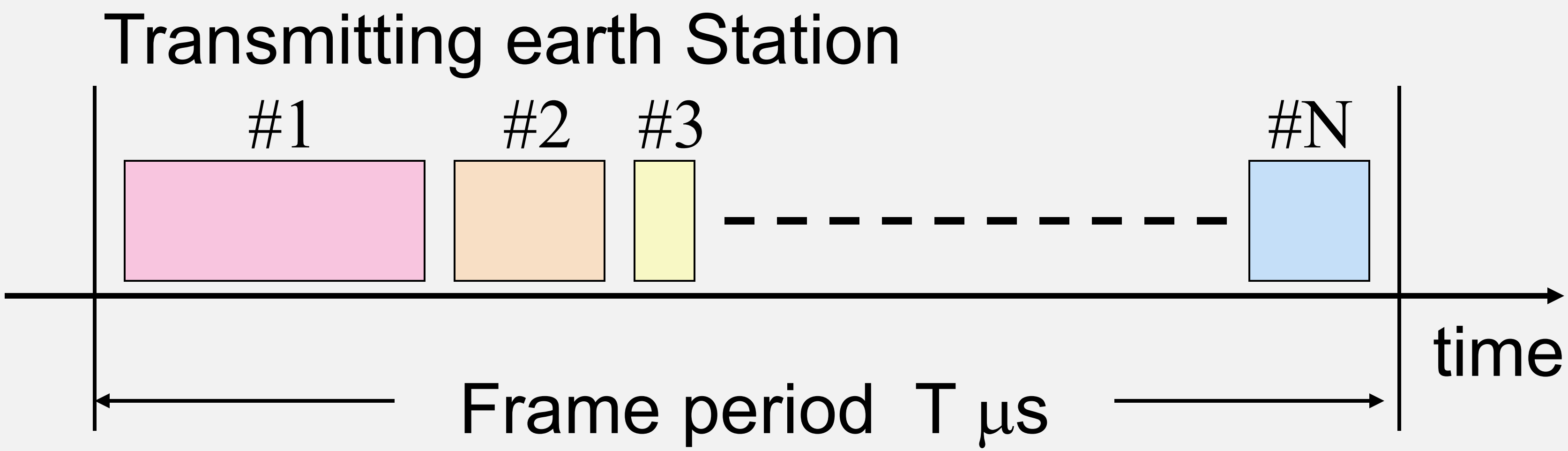
TDMA Basics

Figure 6.20
in text



TDMA Frame Structure

From Figure 6.21 in text



Users occupy a specific time within the frame

Guard times are required between bursts

TDMA Frames

- Frames consist of sequence of bursts from each earth station
- Bursts are completely independent QPSK or QAM transmissions
- Bursts must be separated by guard times to prevent overlap (called *collisions*)
- Some form of start of frame marker is required - could be from control station, or start of station #1 burst

TDMA Frames

- Transmissions are divided into a preamble and traffic bits
- Preamble carries information necessary to identify the bursts and extract their content
- Traffic bits earn revenue while preamble represents overhead

TDMA versus TDM

- Do not confuse TDMA with Time Division Multiplexing (TDM)
- TDM refers to signals being multiplexed in time onto a transmission link at a single location
- TDMA refers to a single transponder being accessed by RF carriers from different Earth stations
- Systems can be structured as TDM-TDMA

TDMAs and Digital Speech

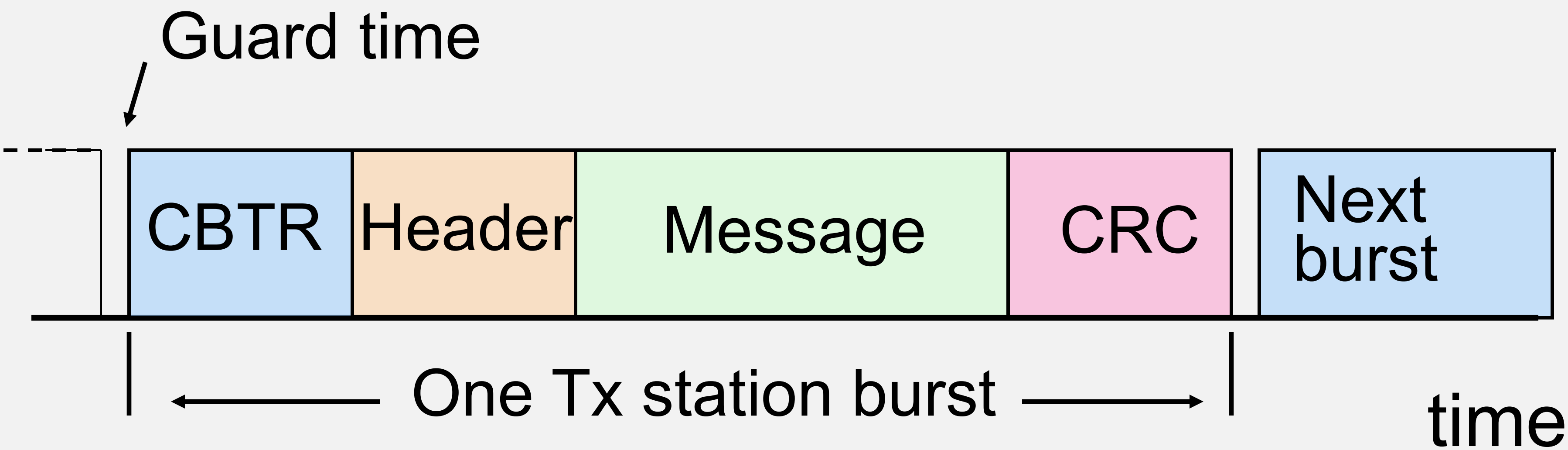
- Early TDMAs were designed with 125 μs frames = 8 kHz repetition rate
- Matches the 8 kHz sampling rate of digital voice as 8-bit digital words
- So a frame transmits one word from each voice channel
- This is sufficient to maintain each voice channel connection in real time

TDMAs and Digital Speech

- The TDMA frame can be lengthened to 2 ms (which is 16x greater)
- Leading to greater efficiency from improved message / preamble ratio
- Now each voice channel will need to deliver sixteen 8-bit words within a frame

TDMA Burst Structure

- Burst looks like a packet with added CTBR



CBTR = carrier and bit timing recovery

Header contains addresses, etc

CRC is cyclic redundancy check

TDMA CBTR

- Each received burst in a TDMA frame is independent
- QPSK demod must lock onto carrier of each burst (carrier recovery to enable bit recovery)
- CBTR is unmodulated carrier + 10101 pattern (Unique Word)
- 176 symbols typical length (3 μs at 60 Msps)
- Demod must synchronize in < 3 μs
- Bit clock must synchronize in < 3 μs

Efficiency in TDMA

- Headers, CBTR, CRC, guard times all waste time that could be used to send data
- Longer frame sends more data, less *overhead*, but adds latency
- Typical TDMA frames are 2 - 20 ms long
- Message bit rate = $R_b \times N / (N + M)$
- where N = number of message bits in frame

  M = total possible bits in frame

Efficiency in TDMA - Example

- Frame length 2.0 ms
- QPSK with symbol rate 50 Msps
- Overhead is 1000 symbols / burst
- Five earth stations with 2 μs guard times
- What is the efficiency of data transfer?

Efficiency in TDMA - Example

- Symbols per frame = $2\ 10^{-3}$ ms x $50\ 10^{6}\ s^{-1}$ = 100,000 symbols
- Overhead per frame = 5 x 1000 = 5,000 symbols
- Guard time per frame = 5 x $2\ 10^{-6}$ ms x $50\ 10^{6}\ s^{-1}$ = 500 symbols
- Lost symbols per frame = 5,500
- Data symbols per frame = 94,500
- Efficiency = 94,500 / 100,000 = 94.5%

TDMA Considerations

- Observe that each burst occupies the <u>full</u> bandwidth of the transponder by itself
- So, less problem with signals interfering in the transponder, i.e., no IM products, less backoff needed

TDMA Considerations

- But the signal is comparatively wideband
- Earth station must transmit at a high bit rate (to occupy full transponder bandwidth)
- Need to operate with higher transmitter power compared to FDMA to obtain same C/N